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PERIOPERATIVE ARRHYTHMIAS

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INTRODUCTION

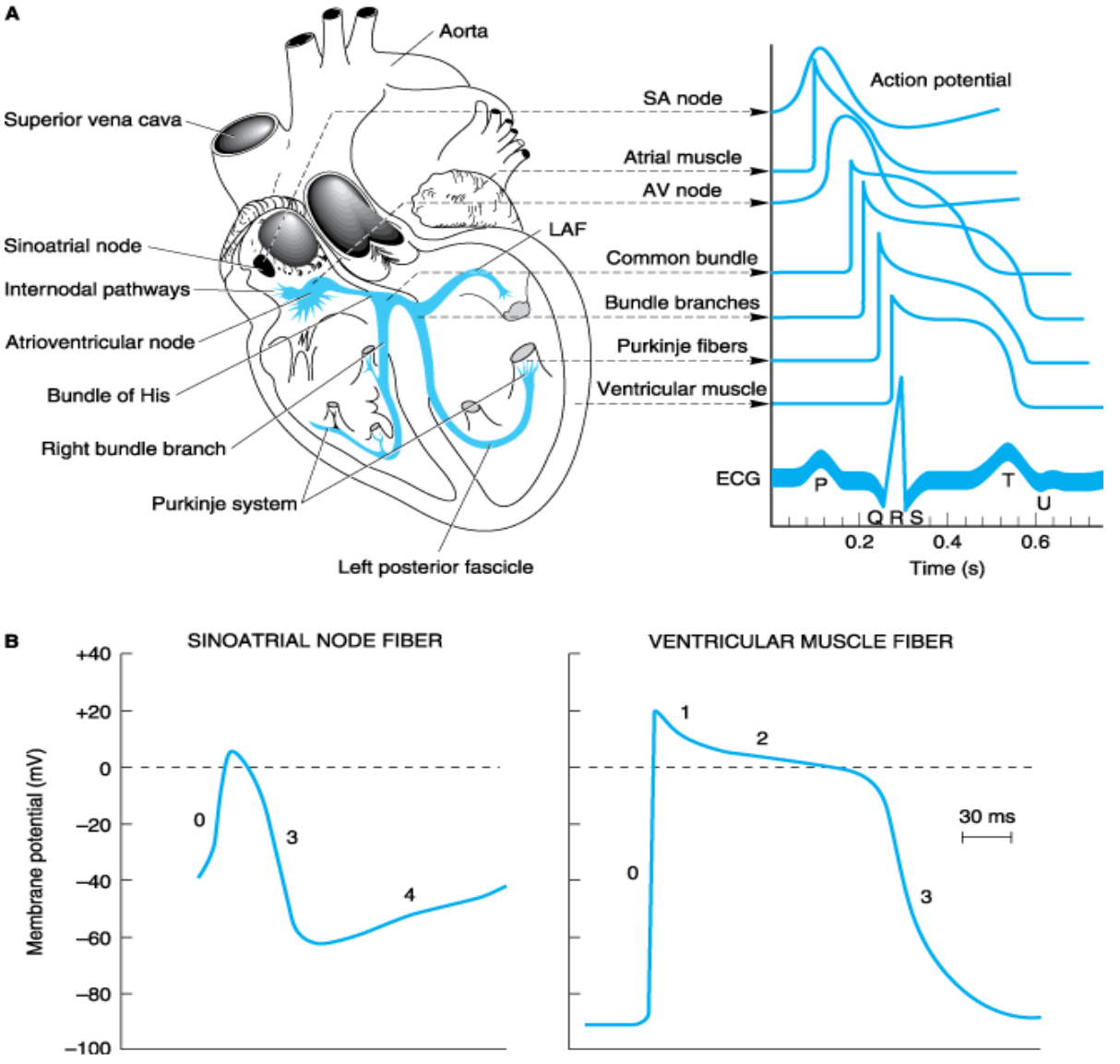
Perioperative cardiac arrhythmias before, during and after anaesthesia range from the innocuous to life threatening. Dysrhythmias are the most frequently occurring perioperative complication.(3) The ability of an anaesthetist to diagnose and treat the arrhythmias is of paramount importance in patient survival and safety as they have the potential to alter the patient's cardiac output and haemodynamic stability. (6) The anaesthetist diagnosing and treating arrhythmias should have a sound approach to ECG interpretation. There are a plethora of weird and wonderful arrhythmias and ECG diagnoses; this booklet will cover the common ones with applicability to the South African public health sector.

ANATOMY, PHYSIOLOGY, ORIGIN OF ARRHYTHMIAS AND MEDICATION ESSENTIAL TO UNDERSTANDING ANTIARRHYTHMIC DRUGS

The Vaughn-Williams (VW) classification presented in 1970, is the most widely recognised classification of antiarrhythmic medications. Since then, various other classification systems have been proposed, including the Sicilian Gambit classification, named after a chess move, which includes drugs not originally included by the VW classification, such as digoxin and adenosine. (7) The intricacies of the pharmacodynamics on individual ion channel mechanisms and complicated channel overlap are beyond the scope of this booklet. The broad classification, main mechanism of action and clinical effects will be covered, as well as clinical use of the medications.

Interestingly, or concerningly, a specialist, quaternary hospital in Kwa-Zulu Natal, has access to merely a paucity of the intravenous medications mentioned in these classification systems**:** lignocaine, adenosine, amiodarone, digoxin and magnesium sulphate. The focus will thus be on these medications.

In order to understand the effects of the medications, the cardiac conduction pathways and physiology must be reviewed. The myocardium is unique as it contains both automatic cells (spontaneous action potentials) and non-automatic cells (induced action potentials). (8)



Source: Morgan GE, Mikhail MS, Murray MJ: *Clinical Anesthesiology*, 4th Edition: http://www.accessmedicine.com

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Figure 1. *A) the ECG waveform is made up from the cumulative electrical activity from all parts of the cardiac excitable tissue. B) the action potentials of the SA node compared to the ventricular myocytes. Image accessed from cvpulmblogspot.com. (9)*

The normal determinants of heart rate are the spontaneous depolarisations of the Sino-atrial (SA) node in the right atrium, slowed down by the parasympathetic nervous system, and increased by the sympathetic nervous system. The SA node is influenced largely by the vagal nerve fibres releasing acetycholine (Ach) onto muscarinic receptors, which cause hyperpolarisation, decreasing the slope of the pacemaker action potential. Similarly, this is seen at the AV node.

The sympathetic nervous system causes release of noradrenaline onto Beta-1 receptors in the pacemaker tissues **and** the myocardium. For the FCA 2 candidate this has relevance for the denervated heart. (10)

The classical ECG waveform that is depicted in textbooks is the sum of the electrical activity occurring at all levels of the myocardium i.e., the SA node and atrial tissue, the AV node and conduction fibres, and the ventricular myocardium.
The automatic action potentials are generated in the SA node and are transmitted to the AV node via the internodal pathways. The electrical impulse is then slowed down in the AV node and transmitted to the His bundle and Purkinje fibres.

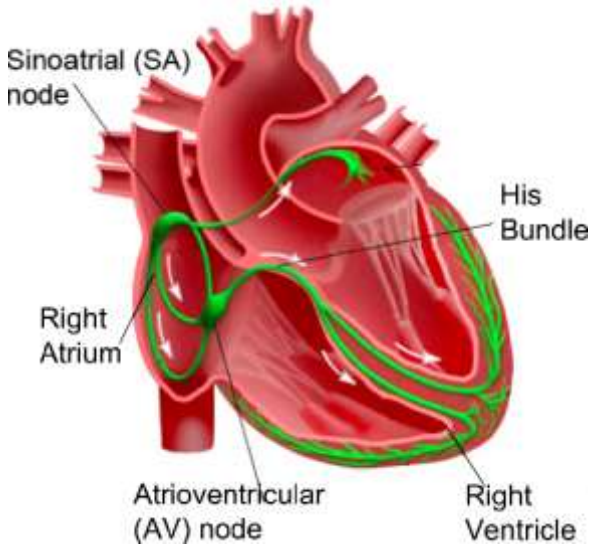


Figure 2. *Cardiac electrical conduction pathways. Image accessed from Simulation of an Artificial Cardiac Pacemaker (11)*

The action potential generation is determined by the inward and outward flow of ions across the pacemaker cell or myocyte. The figures below illustrate the simplified versions of the ion movement.

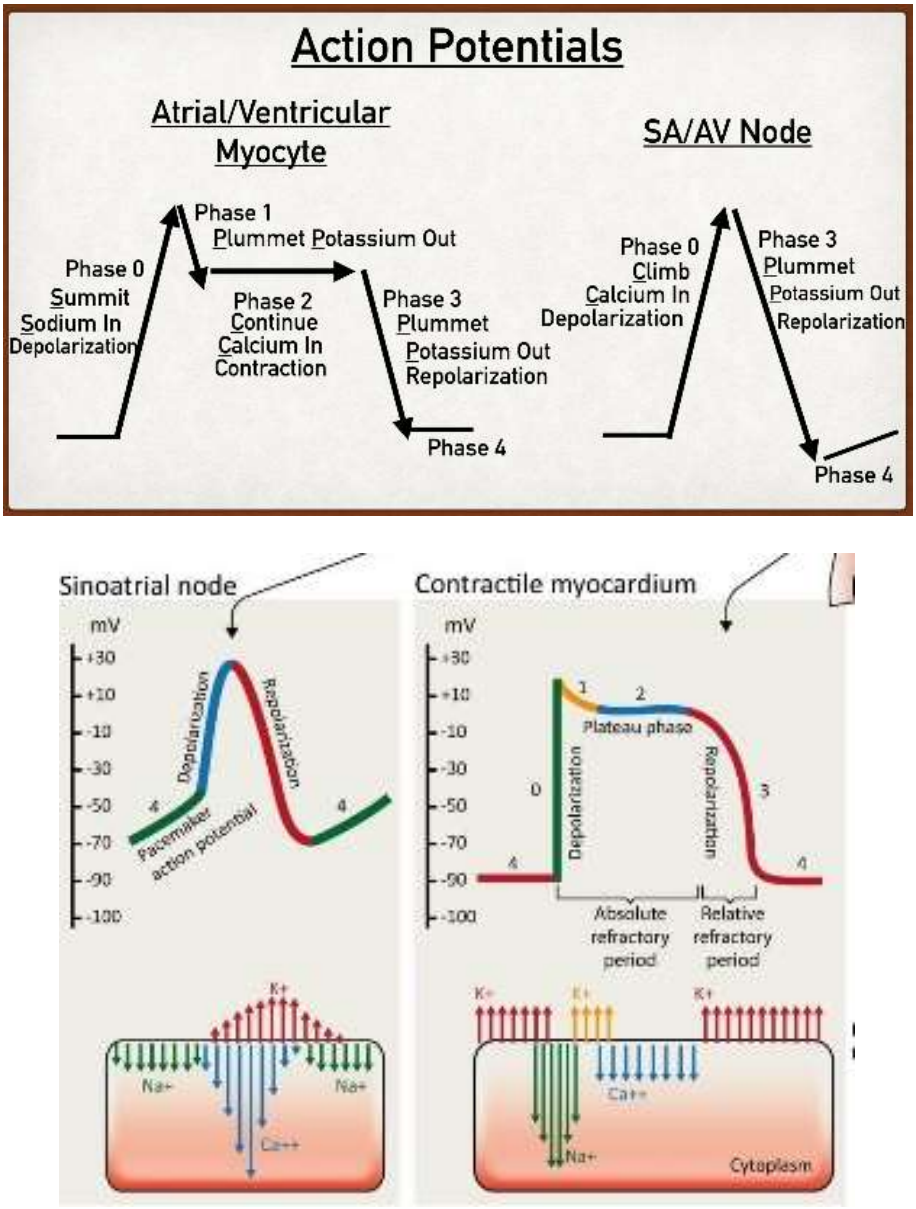


Figure 3. *Simplified ion movement in nodal tissue and myocyte tissue. Images accessed from ezmedlearning.com and ecgwaves.com. (12, 13)*

The Vaughn-Williams classification of antiarrhythmics categorises the main actions of the medications by the predominant ion channel affected. Class 1 drugs affect the sodium channel, Class 2 drugs are beta-blockers, Class 3 affect the potassium channel and Class 4 medications are calcium channel blockers. Drugs not included in this classification system include digoxin, magnesium, and adenosine. Figure 4 below, demonstrates the effect of the antiarrhythmic medications on the nodal and myocyte action potentials.

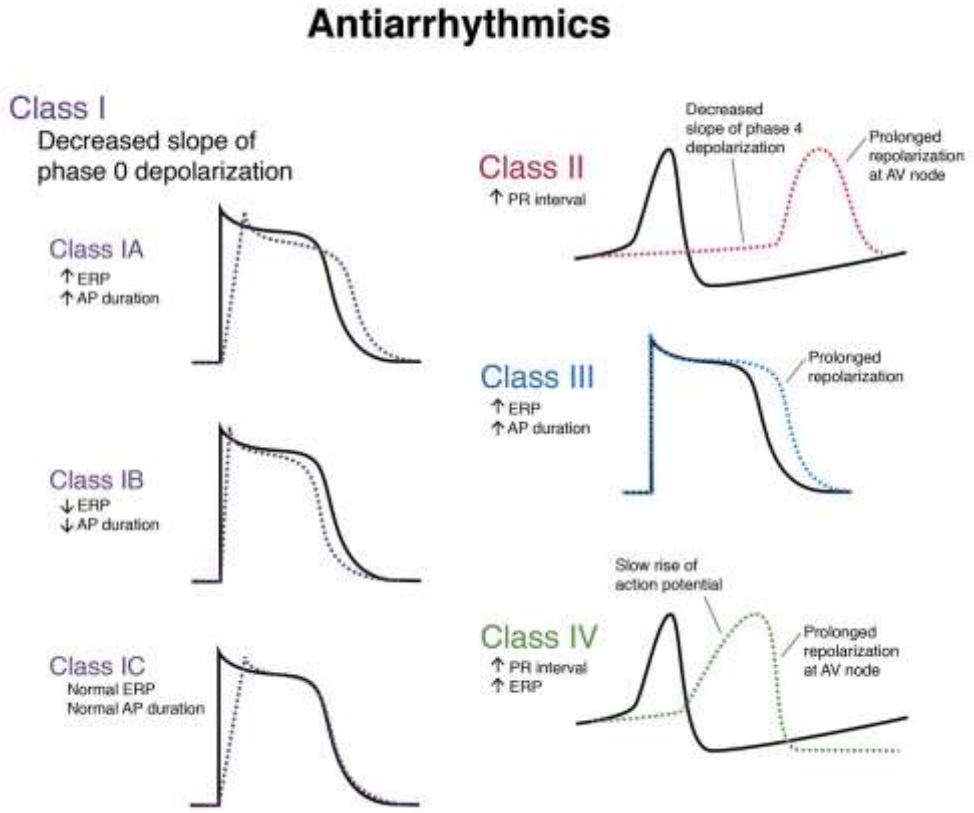


Figure 4. *The Vaughn-Williams drug class effects on the cardiac nodal tissue and myocyte action potentials. Image accessed from medbullets.com (14)*

Class	Examples	Effect on action potential	Ion channels or receptors affected	Used for	Other
I	Na channel blockade				
Ia	Disopyramide, procainamide, quinidine	Slow down phase 0 and lengthen action potential	Sodium channel blockade	Atrial tachycardias, AF, atrial flutter, VT	
Ib	Lidocaine, mexiletene	Slow down phase 0 and shorten action potential	Sodium channel blockade	VT	
Ic	Flecainide, propafenone	Slow down phase 0 and no effect on action potential duration	Sodium channel blockade	Atrial tachycardias, AF, atrial flutter, VT	Avoid in structurally abnormal hearts
II	Beta-blockers except sotalol	Suppress sinus and AV node conduction	Beta-adrenoceptor blockade	Atrial tachycardias, AF, atrial flutter, VT	
III	Amiodarone, dronedarone, sotalol	Lengthen action potential		Atrial tachycardias, AF, atrial flutter, VT	Suitable in structurally abnormal hearts
IV	Verapamil, diltiazem (calcium channel blocker)	Reduce plateau phase and slow AV node conduction	Calcium channel blockade	Atrial flutter, AF, AVNRT	Avoid in WPW syndrome
Other	Digoxin	Slow AV node conduction		Ventricular rate control in atrial flutter, AF, AVNRT	Avoid in WPW syndrome
	Adenosine	Short-acting AV node blockade		Terminating AVNRT, diagnosis of broad-complex tachycardia	Avoid in WPW syndrome and asthma

AF, atrial fibrillation; AVNRT, atrio-ventricular nodal re-entry tachycardia; VT, ventricular tachycardia; WPW, Wolff–Parkinson–White syndrome.

Figure 5. *The Vaughn Williams classification of antiarrhythmics and clinical uses. Image accessed from basicmedicalkey.com. (15)*

Many of the drugs have effects on multiple ion channels which is beyond the scope of this booklet. The illustration below is a simplification of the system with applicability to the South African anaesthetist working in the state sector, particularly in Kwa-Zulu Natal.

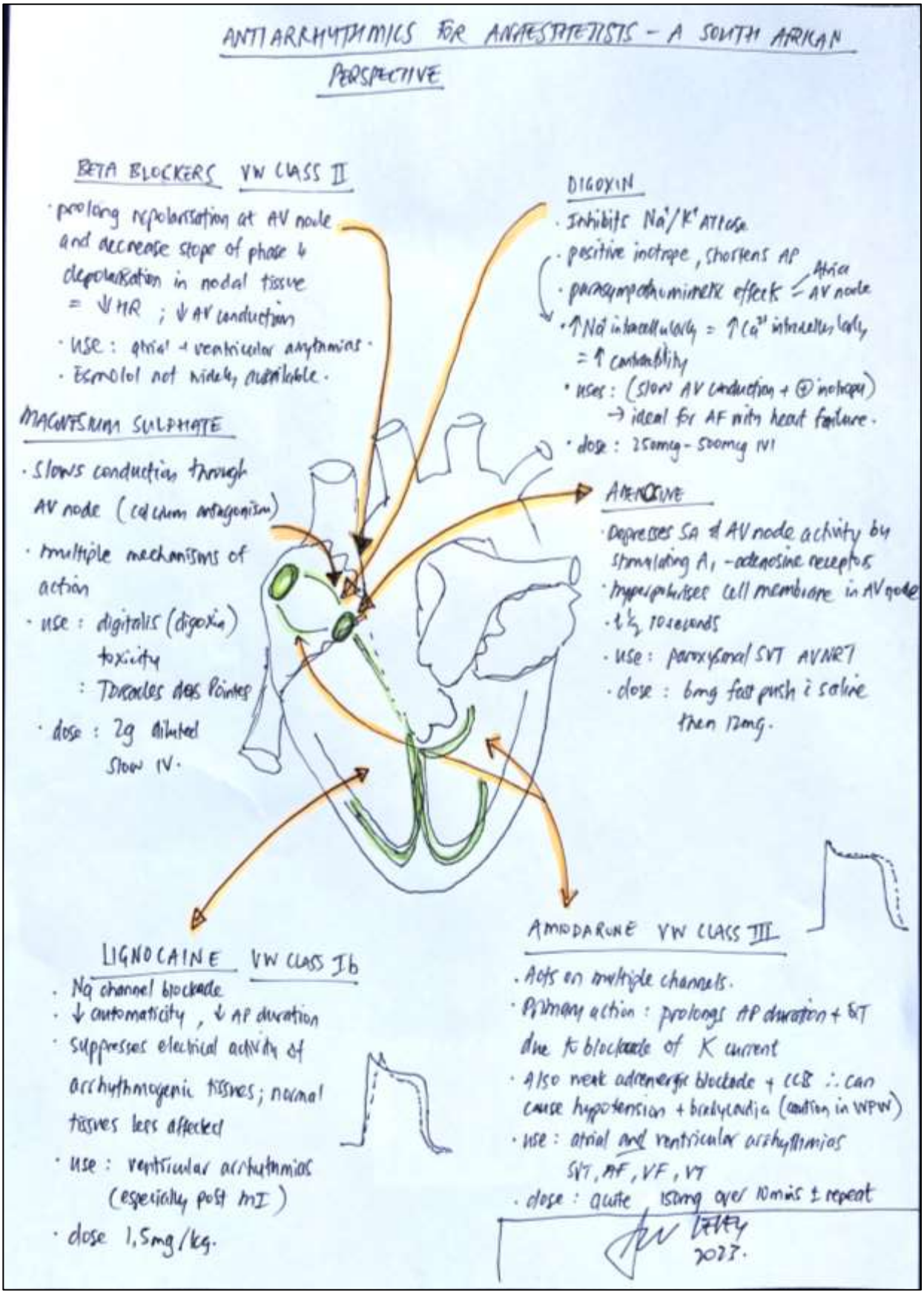


***Figure 6**. The author's interpretation of the commonly encountered anti-arrhythmic agents in the KZN South African public sector. © Andrew M Levey 2023.*

Drugs which need a special mention, not included in the above

- Procainamide – this is a Class 1a antiarrhythmic. It binds to sodium channels and inhibits recovery after repolarisation. The action potential is prolonged and impulse conduction is slowed. It is a unique antiarrhythmic as it blocks abnormal nodal tissue in accessory pathways and is thus used in patients with wide complex ventricular tachycardia, which may be related to accessory pathway activation (see later in booklet) as recommended by the AHA tachycardia with a pulse algorithm. (4) (16)

- Ivabradine: this novel medication selectively inhibits the *funny current* (If) in nodal tissue leading to slowed heart rate (negative chronotropy). Benefit in heart failure with reduced ejection fraction has been demonstrated. As an antiarrhythmic, the drug has been used in patients with inappropriate sinus tachycardias. (17)

PERIOPERATIVE CONTRIBUTORS TO DEVELOPING ARRHYTHMIAS

Broadly, arrhythmias can be caused by

i. Abnormal automaticity of myocardial cells (this may originate in pathological nodal tissue, surrounding nodal tissue or in ventricular myocytes)
ii. Abnormal after depolarisations (an action potential occurs during repolarisation). These can be early or late. The former is classically associated with metabolic and electrolyte disturbances
iii. Re-entry circuits (2)

These underlying mechanisms have precipitants which can be further subclassified according to patient, anaesthetic and surgical factors; these will be discussed below.

Patient factors: overview

The incidence of perioperative arrhythmias can vary, depending upon the patient's underlying cardiovascular comorbidities, their various chronic medications and their preoperative optimization. The abnormal arrythmias may also be related to the physiological effects of the anaesthetic technique used in the individual patient.(6) (18)
A pre-operative ECG is generally attained in the elderly or patients with comorbidities contributing to cardiovascular disease such as hypertension, diabetes, and dyslipidaemia. Patients with certain underlying diseases are more prone to developing arrhythmias e.g., patients with mitral valve disease are prone to developing atrial fibrillation.
In healthy patients, it may be useful to compare a pre-operative ECG to an intra- or post- operative ECG. (19) A systematic approach should be utilised when interrogating ECG's. Specific underlying genetic or acquired conditions predisposing patients to developing arrhythmias should be sought.

Figure 1 below, reminds the reader of the normal ECG curve, waveforms, and segments.

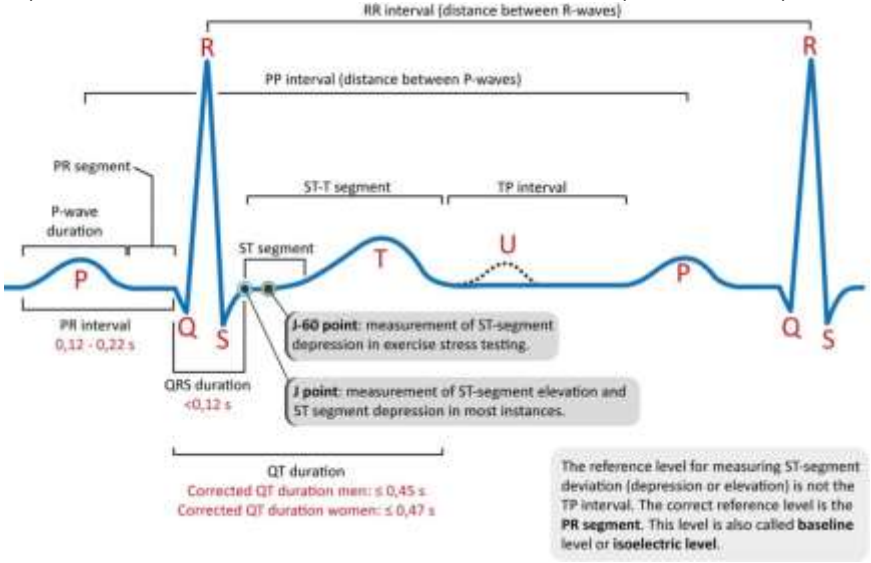


Figure 7. *ECG waveform and segments, accessed from www.ecgwaves.com (2)*

Patient factors: pre-operative ECG

Pre-operative ECG's equip clinicians with an opportunity to identify potential arrhythmogenic diseases or disorders. Potential arrhythmogenic disorders can be identified by systematic analysis of the ECG as follows:

Rate: Tachyarrhythmias and bradyarrhythmias – discussed below

P waves: the presence of enlarged P waves (P Pulmonale or P Mitrale) suggests atrial enlargement, which predisposes the patient to atrial arrhythmias (discussed below). Underlying conditions associated with the atrial enlargement should be sought e.g., mitral stenosis, chronic lung disease.

PR interval: normal 120-200msec. Pre-excitation syndromes have the potential to escalate into unstable tachyarrhythmias**.** The treatment for these syndromes must avoid nodal blocking agents as these may precipitate rapid conduction of the supraventricular tachycardia (SVT) down the accessory pathway (see AV nodal re-entry tachycardia AVRT below). There are two types of pre-excitation syndromes with which the anaesthetist should be familiar

Wolff Parkinson White

In this condition some of the impulses generated in the SA node are transmitted via an accessory pathway and not via the AV node, causing a short PR interval and the characteristic delta wave. The remainder of the SA node impulses traverse the AV node; this normal conduction with the delta wave can cause a prolonged QRS. However, it will be <110ms as opposed to the lengthened duration seen in a bundle branch block. Patients are prone to developing tachyarrhythmias.(20)

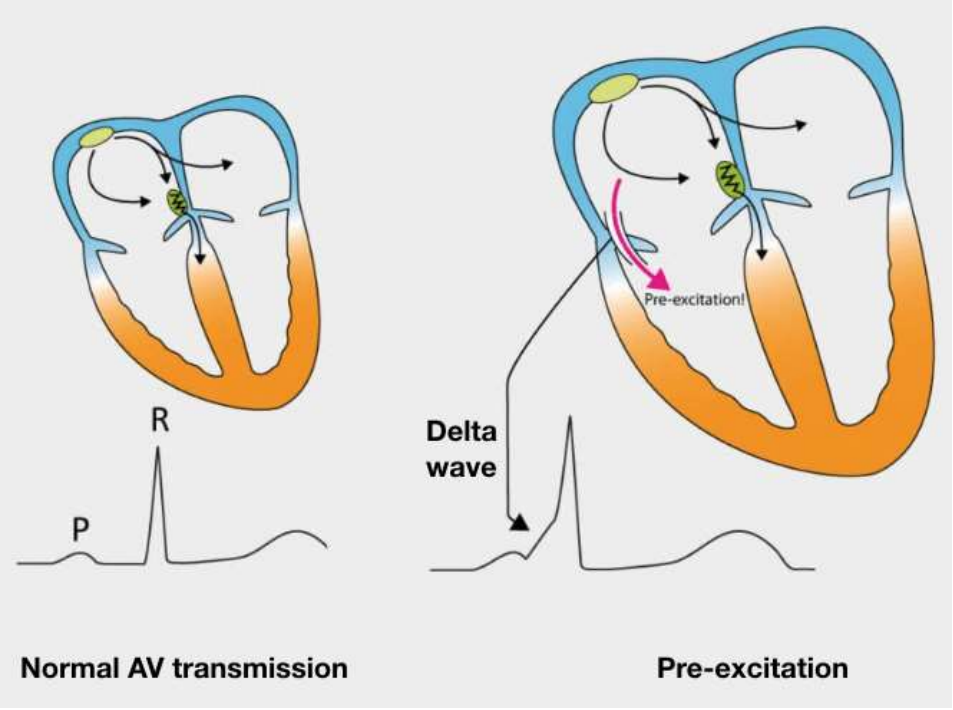


***Figure 8**. Accessory pathway conduction causing a delta wave in WPW. Image accessed from Life in the Fastlane. (20)*

Type A WPW has the SA nodal action potential conducted via the left side of the heart producing a positive delta wave in all precordial leads; this type can mimic RVH and pseudoinfarction in aVL.

Type B WPW is conducted via the right side of the heart and produces a negative delta wave in V1. (20)



***Figure 9.** Type A, "left sided" WPW, positive R and delta wave in V1. Image accessed from Life In the Fastlane (20)*



Figure 10. Type B, "right sided" WPW, negative delta wave in V1 with S wave. Image accessed from Life in the Fastlane. (20)

The other pre-excitation syndrome, known as the Lown-Ganong-Levine syndrome (LGL) number

This has a short PR interval with no delta wave. There is controversy over whether this entity actually exists. (20)

Q waves: when present, prior myocardial infarction should be suspected: this predisposes the patient to arrhythmias and conduction blocks.

QRS: Prolongation = >120msec.

QRS duration: Left bundle branch block (LBBB) or right bundle branch block (RBBB)

Bundle branch blocks with a QRS duration of <120msec are termed *incomplete BBBs*. These remain significant as they can progress to complete BBB's. (2) The bundle branches are depicted in Figure 11 below.

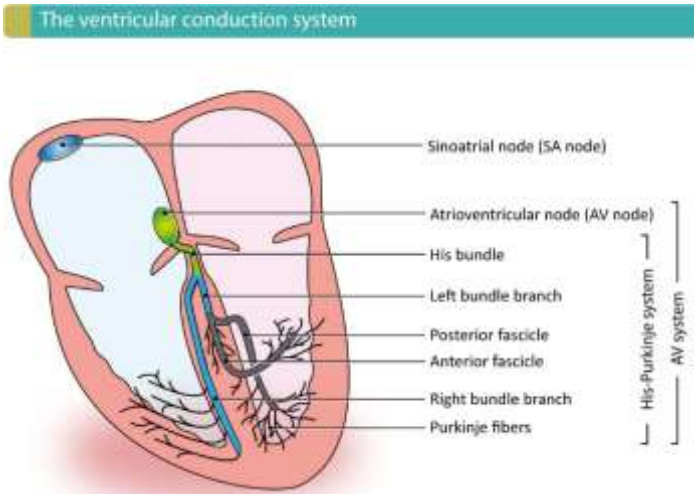


Figure 11. *Ventricular conduction system. Image accessed from ECG & ECHO waves (2)*

QRS LBBB: the implication of a left bundle branch block is that this may indicate underlying conduction defects or, if acute, may indicate new onset ischaemia. A LBBB is diagnosed by identifying a wide QRS with a rS pattern in V1/V2 (initial r wave and deep S wave) and broad R wave, which may be notched, in V5/V6. (21). An easy mnemonic method of remembering the morphology is WILLIAM, where a "W" pattern is seen in V1/2 the "LL" reminds one that this relates to a Left BBB and the "M" is the pattern seen inV5/6. (See fig 3). For the interested reader: suggested further reading, Sgarbossa criteria in LBBB with suspected STEMI.

QRS RBBB: QRS > 120msec; V1/2 rsR / rsR / rsr pattern (the second R wave is taller than the first); V5/6 qRs pattern with slurred s wave. S wave is wider and smaller than R wave. (21). The mnemonic MORROW can be used here to aid memory. A new onset RBBB may indicate pathology such as an acute pulmonary embolism. (19) Importantly, RBBB may be benign or indicate a severe conduction disorder. Right bundle branch blocks do usually not alter the electrical axis of the heart. Thus, in the presence of axis deviation, this indicates that associated fascicular blocks are present.(2) ST segment deviation from the baseline is easier to interpret in RBBB than in LBBB as the majority of ischaemia/infarctions occur on the left side of the heart. Figure 12 below illustrates the morphology of a RBBB vs a LBBB.

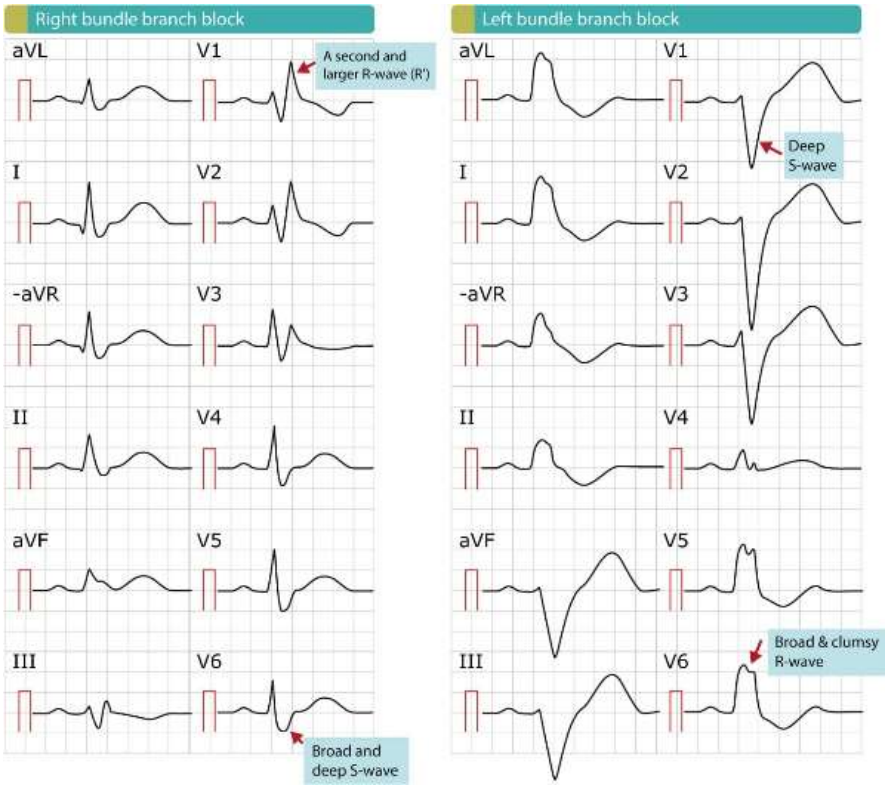


Figure 12. *RBBB and LBBB: image accessed from ECG & ECHO waves (2)*

Causes of RBBB and LBBB should always be sought. Examples are presented in Table 1.

RBBB	LBBB
Normal variant	Hypertension
Secondary to lung disease: • Cor pulmonale • Pulmonary embolism	• Aortic stenosis • aortic root dilatation
Ischaemic heart disease	
Electrolyte abnormalities	DCMO
Congenital heart disease	Degeneration of electrical conduction system
Lev's disease	

Table 1: *causes of RBBB and LBBB.*

Caution: Patients with a LBBB are at risk of developing a complete heart block with insertion of right sided venous catheters. (19) They should thus have external cardiac pacing facilities available on insertion.

The development of a new, transient BBB may be caused by tachycardias. This is also termed *rate-dependent bundle branch block* or, *tachycardia related BBB*. (22)

QRS continued…A wide QRS may also be seen after an ischaemic event such as STEMI or post cardiac bypass. This is termed *Accelerated Idioventricular Rhythm or* ***AIVR*** and is illustrated in Figure 13 below. This occurs when there are ventricular pacemaker foci depolarising at a faster rate than the sinus node. There is usually not an associated tachycardia. Hallmarks of the AIVR are a prolonged QRS; three or more ventricular complexes; a regular rhythm and fusion as well as capture beats. The arrhythmia is usually self-limiting and causes minimal haemodynamic issues. Treatment should be aimed at correcting the underlying cause. (23)

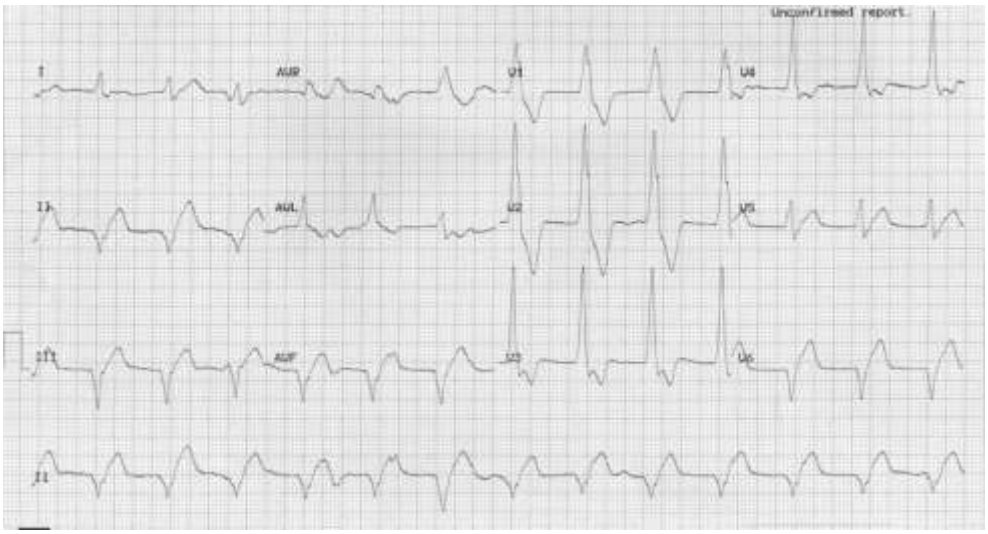


Figure 13: *ECG demonstrating ventricular, wide QRS seen in AIVR. A fusion beat can be seen in standard lead 2. Image accessed from Life in the Fastlane. (23)*

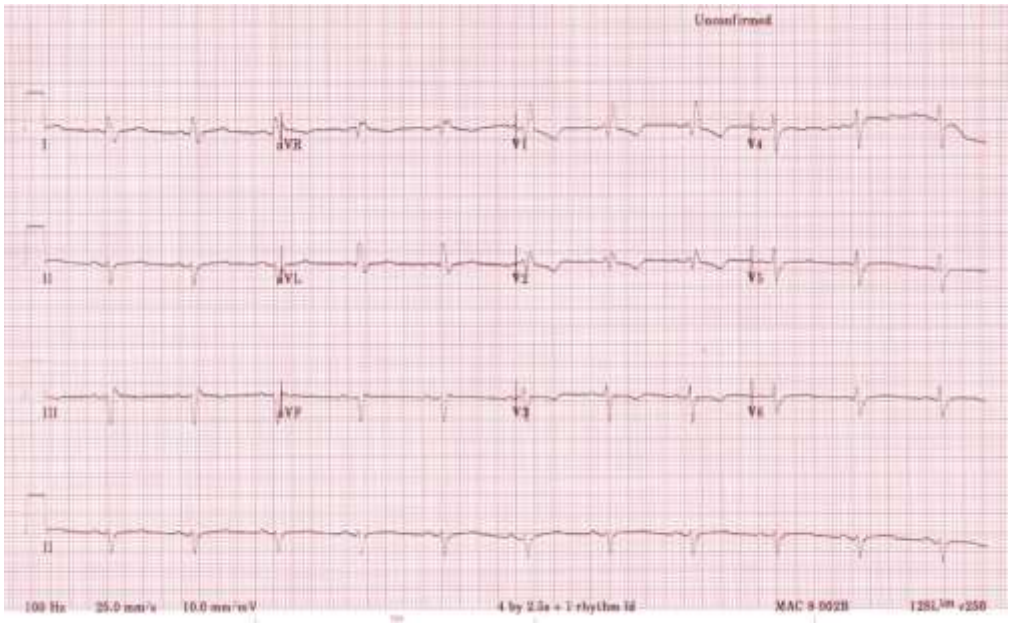


Figure 14: *RBBB with left axis deviation; left anterior fascicle blocks. Image accessed from Life in the Fastlane. (24)*

QRS: Fascicular blocks

Bifascicular blocks - RBBBs in the presence of axis deviation indicate fascicular blocks.

RBBB with left axis deviation indicates a RBBB and left anterior fascicular block. (24). This indicates that there is single coronary artery blood supply to the anterior fascicle (left anterior descending coronary artery). (24)

RBBB with right axis deviation indicates RBBB and left posterior fascicle block. Conduction system and structural heart disease is present in half to three quarters of patients with bifascicular blocks; importantly there is a risk of progression to complete heart block if the remaining fascicle is blocked. Useful clinical interrogation in the presence of bifascicular blocks is to ask the patient if there is a history of syncope; a positive answer implies risk of progression to complete heart block. (24)

The presence of chest pain with a bifascicular block has a strong association with proximal LAD occlusion. (24)

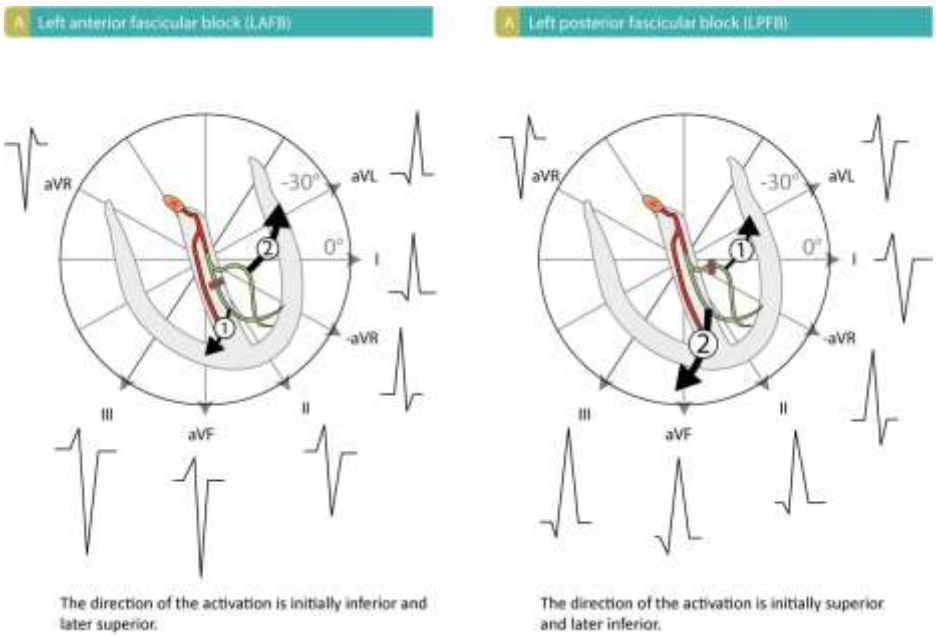


Figure 15: *LAFB left axis deviation due to left anterior fascicle block; LPFB right axis deviation due to left posterior fascicle block. Image accessed from ecgwaves.com. (2)*

QRS: Trifascicular blocks

True trifascicular blocks indicate the blockage of three fascicles e.g., the left bundles anterior and posterior fascicles and the right bundle. This demonstrates severe conduction disease and, if identified pre-operatively, warrants referral to cardiology and possible pacemaker insertion. The block is identified by:

i. Right bundle branch block
ii. Axis deviation
iii. Complete heart block (25)

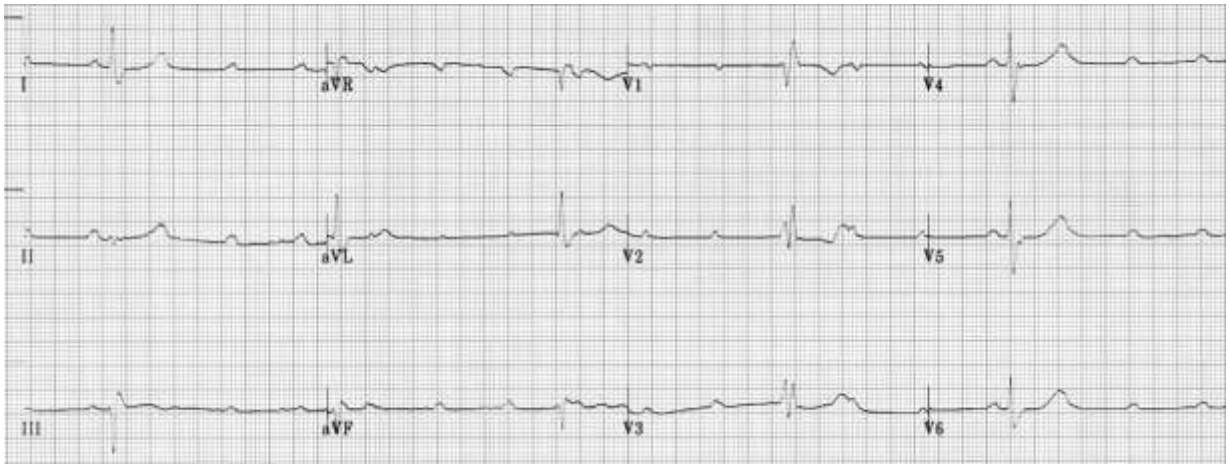


Figure 14. *Image accessed from Life In The Fastlane demonstrating a true trifascicular block with i) complete heart block, ii) RBBB, iii) axis deviation to the left (LAFB). (25)*

QRS: Incomplete trifascicular blocks also exist, which additionally demonstrate serious degeneration of the conduction system; referral prior to elective surgery is prudent.

The block consists of the triad of

i. RBBB
ii. Axis deviation (fascicle block)
iii. AV node delay as demonstrated by prolonged PR interval (25)

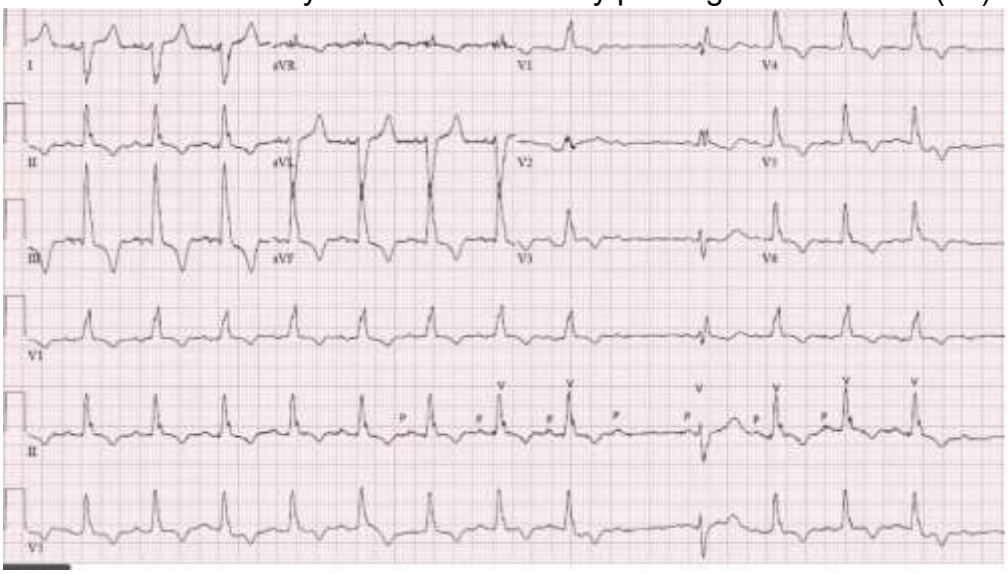


Figure 15. *Incomplete trifascicular block as demonstrated by i) RBBB, ii) right axis deviation (LPFB), iii) PR prolongation with dropped beat (Mobitz II heart block). Image accessed from Cureus. (26)*

QRS: Axis deviation and calculation

Table 2 below presents the causes of left and right axis deviation.

Left axis deviation	**Right axis deviation**
LVH	RVH
Inferior wall MI	Lateral wall MI
LBBB	RBBB
LAFB	LPFB
WPW	
	Pulmonary hypertension
	Dextrocardia

Table 2: *causes of left and right axis deviation. (21)*

QRS: **how to accurately calculate the axis**

There are many methods of calculating the electrical axis on an ECG. The author finds the most accurate method to calculate the mean electrical axis (MEA) is by using an intersecting line between two vectors. The net deflection in lead 1 and the net deflection in aVF are drawn on to the circle of axes. The mean electrical axis is the vector drawn between the two. (27) The following normal ECG (figure 9, below) will be used as an example:

To calculate the MEA: Lead 1: + 12 aVF: -1 + 11 = 10. MEA drawn as arrow intersecting vectors in I and aVF. When applied to system of axes this approximates to approximately 45', which is a normal axis.

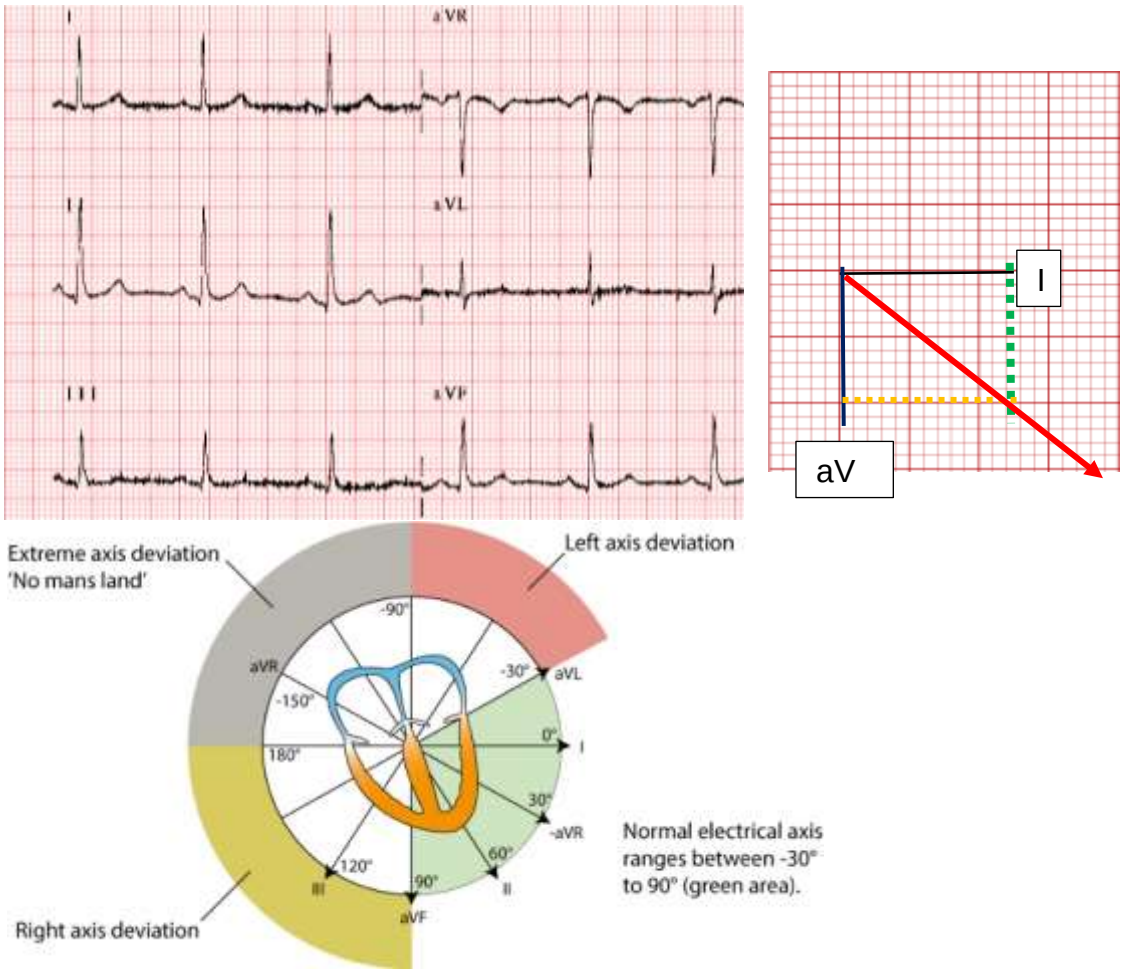


Figure 16: *Normal ECG, system of axes and mean electrical axis. Image accessed from ekgecho.de (28)*

Pre-operative ECG continued
ST segment: Brugada syndrome

Brugada syndrome is an inherited channelopathy (50% spontaneous mutations, some autosomal dominant inheritance pattern, sixty different mutations described thus far). These patients have structurally normal hearts. They are at risk of VT and VF. The incidence is 1 in 1000 and most prevalent in Asian males. (29) The channelopathy, involving dysfunctional cardiac sodium channels, leads to early loss of the plateau of the action potential. The diagnosis is made with ECG findings and clinical criteria including documented VT or VF, positive family history of sudden cardiac death, inducible VT, and the like. The definitive treatment is an implantable cardiac defibrillator. (30)

Figure 18 below demonstrates the different types of Brugada syndrome. Figure 19 demonstrates the coved type pattern. Figure 20 demonstrates the relationship between the transmembrane action potentials and the surface ECG.



Figure 17: *Three types of Brugada syndrome. Image accessed from Life in the Fastlane. (30)*

Type 1: Coved ST segment elevation >2mm in 2 or more of V1-V3 followed by a negative T wave. This is also known as *the Brugada Sign.*
Type 2: Saddleback ST elevation >2mm in V1-3
Type 3: ST elevation in V1-V3 (saddle or coved) <2mm (30)

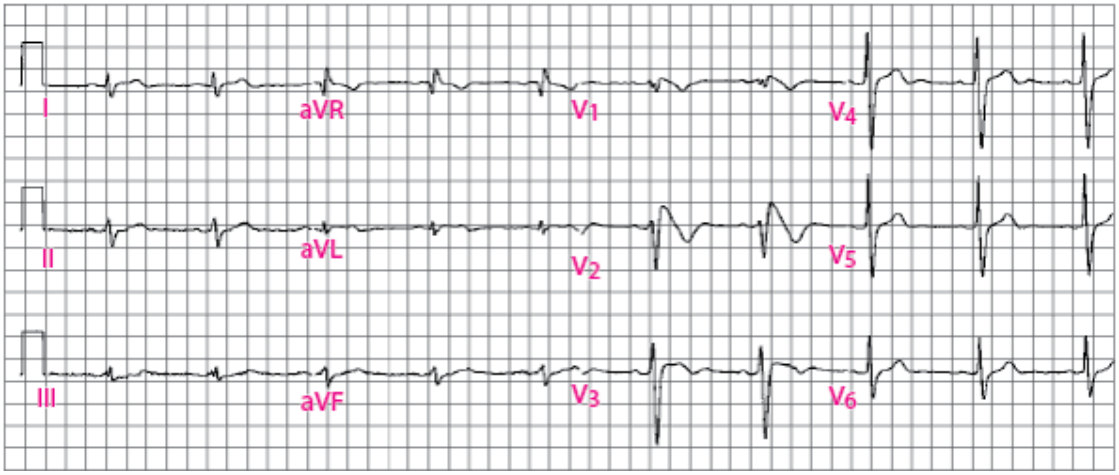


Figure 18: *Brugada Type 1 Coved ECG pattern. Image accessed from www.msdmanuals.com. (29)*

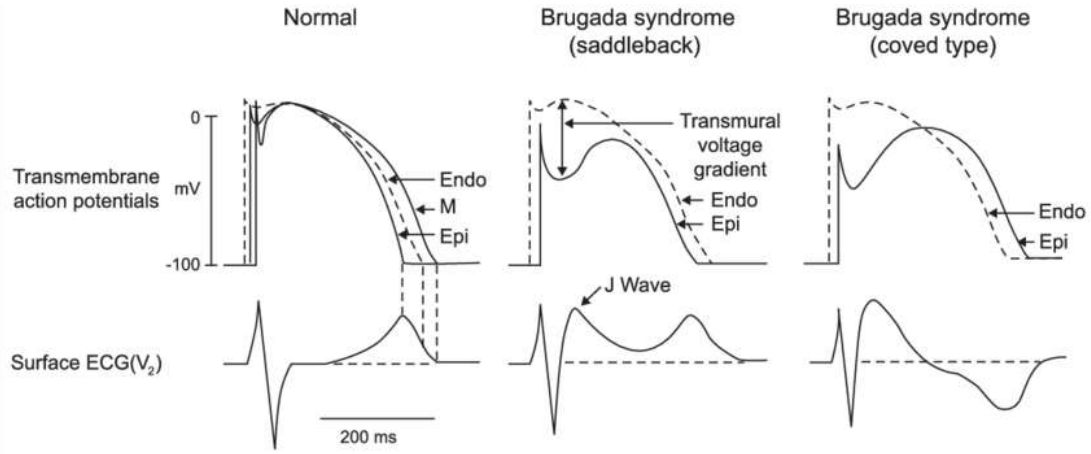


Figure 19: *Action potentials in Brugada syndrome demonstrating the saddleback and coved type. Image accessed from www.researchgate.net. (31)*

ST segment continued

Interested readers are encouraged to familiarise themselves with mimickers of STEMI's, including LV aneurysms, benign early repolarisation, coronary spasm, pericarditis, paced rhythms amongst others.

ST elevation and depression

The diagnosis of intra-operative ST elevation myocardial infarction and non-ST elevation myocardial infarctions are deemed essential knowledge to the anaesthetist and commonly cause arrhythmias, notably inferior myocardial infarctions can cause nodal dysfunction, hypotension and heart blocks. This booklet will not cover these.

Other arrhythmogenic preoperative ECG findings: Dilated cardiomyopathy, hypertrophic cardiomyopathy

Dilated cardiomyopathy (DCM)

DCM is a specific term related to an idiopathic or genetic dilatation of the left ventricle. Other causes of cardiomyopathies may lead to the same structural changes as DCM however they have a defined, underlying cause – examples include post-partum dilated cardiomyopathy, tachycardia induced cardiomyopathy and related and ischaemic cardiomyopathy. If advanced LV involvement is present these patients are prone to developing ventricular arrhythmias leading to sudden death. (32) Fig 21 demonstrates non-specific intraventricular conduction delay and T wave inversion.

Typical ECG findings include:

Signs of LV involvement: LBBB, LVH, non-specific ST & T wave changes

- Intraventricular conduction delays which are non-specific. (QRS > 100msec)

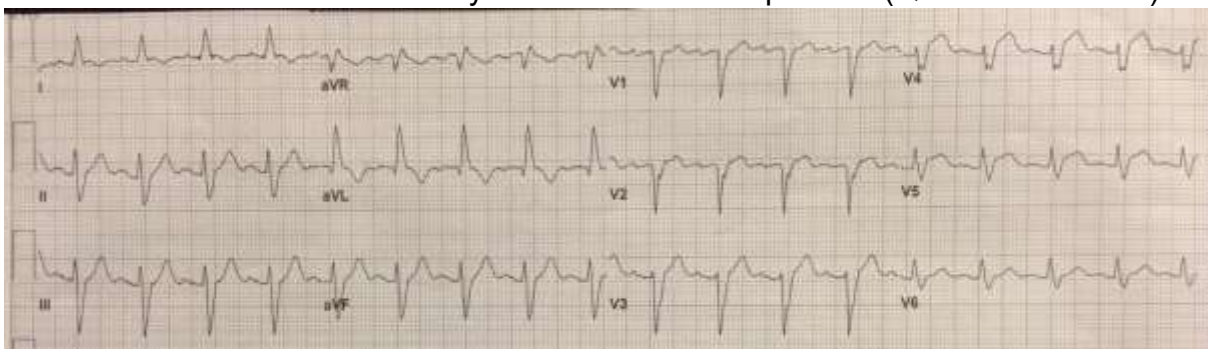


Figure 20 *demonstrating non-specific intraventricular conduction delay and T wave inversion. Image accessed from hqmeded-ecg.blogspot.com. (33)*

Hypertrophic cardiomyopathy

A preoperative ECG may reveal features of LVH with associated deep and narrow Q waves in the lateral and inferior leads. These are commonly referred to as “dagger” Q waves; morphologically these are seen as being <40ms in width (interestingly STEMI q waves are >40ms in width). This disease is mentioned as the patients are prone to developing SVT’s and atrial fibrillation. (34) Classical HOCM ECG showing dagger like Q waves in inferior leads and lateral leads; voltage criteria for LVH are presented in figure 22 below.

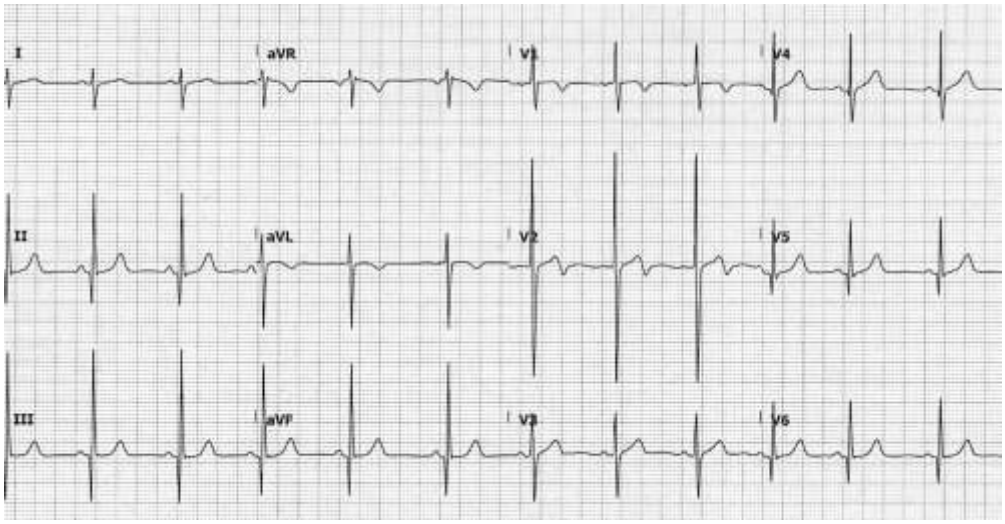


Figure 21*: Classical HOCM ECG showing dagger like Q waves in inferior leads and lateral leads; voltage criteria for LVH. Image accessed from Life in the Fastlane. (34)*

Arrhythmogenic conditions continued

Long QT syndrome (LQTS)

Prolongation of the QT interval can be caused by electrolyte disturbances, myocardial infarctions, rheumatic heart disease, medication related, or can be congenital. Electrolyte disturbances predisposing to a long QT interval include hypomagnesaemia, hypocalcaemia and hypokalaemia. The normal QT interval for males is <440ms and <460ms for females. Increased heart rates decrease the QT

interval, whereas slower heart rates increase the QT interval. The corrected QT interval (QTc) is adjusted to a heart rate of 60 bpm. There are multiple complicated formulae to calculate the QTc including Bazett's and Framingham's formula. (35)

Clinically, the anaesthetist can examine the distance between two RR intervals. If the T wave ends more than halfway between the two intervals, the QT interval is prolonged. Peri-operatively, patients known with prolonged QT interval should have a baseline ECG and be discussed with cardiology. An implantable cardiac defibrillator device may be required in severely prolonged QT intervals or in patients who have developed previous lethal arrhythmias. It is suggested that drugs that can prolong the QT interval are avoided (see figure 23) Tight control of electrolytes should be instituted pre- and intra-op. *Propanolol* is a beta blocker commonly used to treat patients with congenital LQTS (CLQTS); this should be continued peri-operatively if the patient has been known to be taking it chronically.

Patients with CLQTS are at risk of developing *Torsades de* Pointes and ventricular fibrillation.

The peri-operative period predisposes the patient with CLQTS to multiple potential pro-arrhythmic sources including catecholamines as well as metabolic and electrolyte disturbances. Premedication with benzodiazepines can assist with anxiolysis as well as general noise reduction.

Standard anaesthetic monitoring is appropriate with special considerations for a 5 lead ECG, defibrillator in theatre, readily available magnesium, and possible placement of transthoracic pacing pads. The anaesthetist should aim for a cardiovascularly stable anaesthetic and maintain normal metabolic and electrolyte parameters. While propofol and opioids are considered safe for induction, ketamine is best avoided. Rocuronium and cisatracurium are deemed safe. Suxamethonium should be avoided. Caution should be used using anticholinesterases and sugammadex is preferred, if available. Volatile anaesthetics can potentially lengthen the QT interval and thus propofol TIVA may be a more prudent option. The clinical significance of using volatile anaesthesia is, however, probably insignificant. Epidural anaesthesia without adrenaline as an additive, is preferred to spinal anaesthesia which can cause rapid haemodynamic changes. (36)

Caution should be exercised when administering PONV prophylaxis as ondansetron and droperidol can lengthen the QT interval.

Should Torsades de Pointes (TdP) or ventricular fibrillation develop intra-operatively the following treatment algorithm is suggested:

Table 3. *Treatment of Torsades des Pointes. (18)*

Torsades de Pointes treatment (18)	
Haemodynamically stable	Haemodynamically unstable
Slow loading Magnesium 2g IVI	Synchronised cardioversion and load with magnesium
Isoprenaline or overdrive pacing to decrease QT interval	Pulseless: CPR and defibrillation, magnesium loading and consider lignocaine

Table 4 demonstrates risk factors in developing TdP. (36)

Factor	Condition or Trigger
Metabolic factors ("hypos")	Hypokalaemia, hypomagnesaemia, hypocalcaemia, hypothyroidism, hypothermia
Bradyarrhythmias	Sinus node dysfunction, atrioventricular block
Other clinical factors	Female sex, myocardial ischemia, intracranial disease, HIV, autoimmune diseases
Androgen deprivation	Bilateral orchiectomy, leuprorelin
Antianginal drugs	Ivabradine, ranolazine
Antiarrhythmic drugs	Disopyramide, hydroquinidine, procainamide, quinidine, flecainide, pilsicainide, propafenone, amiodarone, dofetilide, dronedarone, ibutilide, nifekalant, sotalol, vernakalant
Anticholinergic drugs	Solifenacin, tolterodine
Anti-infective drugs	
Fluoroquinolones	Ciprofloxacin, gatifloxacin, gemifloxacin, grepafloxacin, levofloxacin, moxifloxacin, sparfloxacin
Macrolide antibiotics	Azithromycin, clarithromycin, erythromycin, roxithromycin
Antituberculous drugs	Bedaquiline
Azole antifungals	Gluconazole, voriconazole
Antiparasitic drugs	Pentamidine
Antimalarial drugs	Chloroquine, delamanid, halofantrine, hydroxychloroquine, quinine, piperaquine
Antiretrovirals	Saquinavir
Antihistamines	Astemizole, hydroxyzine, terfenadine
Antineoplastic drugs	Aclarubicin, arsenic trioxide, ceritinib, crizotinib, dasatinib, encorafenib, gilteritinib, inotuzumab ozogamicin, ivosidenib, lenvatinib, midostaurin, nilotinib, osimertinib, oxaliplatin, ribociclib, toremifene, vandetanib, vemurafenib
Diuretics	Via electrolyte changes
Gastrointestinal drugs	
Antimemetics	Droperidol, ondansetron
Promotility drugs	Cisapride, domperidone
Neuromuscular blockers	Succinylcholine, pancuronium
Neurologic drugs	Apomorphine, deutetrabenazine, donepezil, ezogabine, fingolimod, pimavanserin, tetrabenazine
Opioids	Buprenorphine, hydrocodone, levacetylmethadol, methadone, sufentanil
Psychotropic drugs	
Antipsychotics	Amisulpride, chlorpromazine, chlorprothixene, clozapine, flupentixol, haloperidol, levomepromazine, mesoridazine, olanzapine, pimozide, quetiapine, risperidone, sulpiride, thioridazine, ziprasidone
Antidepressants	Citalopram, clomipramine, doxepin, escitalopram, imipramine
Other	Atomoxetine, trazodone, valbenazine
Anaesthetics and sedatives	Chloral hydrate, ketamine, propofol, thiopentone, desflurane
Vasodilator drugs	Bepridil, cilostazol, intracoronary papaverine
Miscellaneous	Anagrelide, alfuzosin, cesium chloride, clofazimine, cocaine, eliglustat, ibogaine, lofexidine, mifepristone, oxytocin, pasireotide, probucol, terlipressin, terodiline, organophosphate intoxication

Table 4: *Risk factors in congenital LQTS, accessed from World Federation of Societies of Anaesthesiologists, Long QT Syndrome and Anaesthesia (36)*

Arrhythmogenic conditions continued

T waves

T waves represent ventricular repolarisation. They are generally upright in all leads except aVR and V1. Their amplitude is 0.5cm in limb leads and <1cm in precordial leads. Hyperacute T waves are seen in STEMI and Prinzmetal angina.

Inverted T waves can represent myocardial ischaemia (discussed below), represent ventricular hypertrophy or be part of bundle branch blocks; they are also seen with HCM, pulmonary embolism and raised intracranial pressure. They are a normal finding in children or may represent a persistent juvenile T wave pattern. (37)

New T wave inversion is always a cause for concern, ischaemia should be excluded. T wave inversion with associated Q waves can represent prior myocardial infarction.
Wellen's syndrome represents critical ischaemia in the left anterior descending artery. There are two types:

- Biphasic T waves in V2 and V3 represent **type A** and occur in 25% of cases.
- Deep, symmetrically inverted T waves in V2 and V3 occur in 75% of cases, known as **type B**. (37)

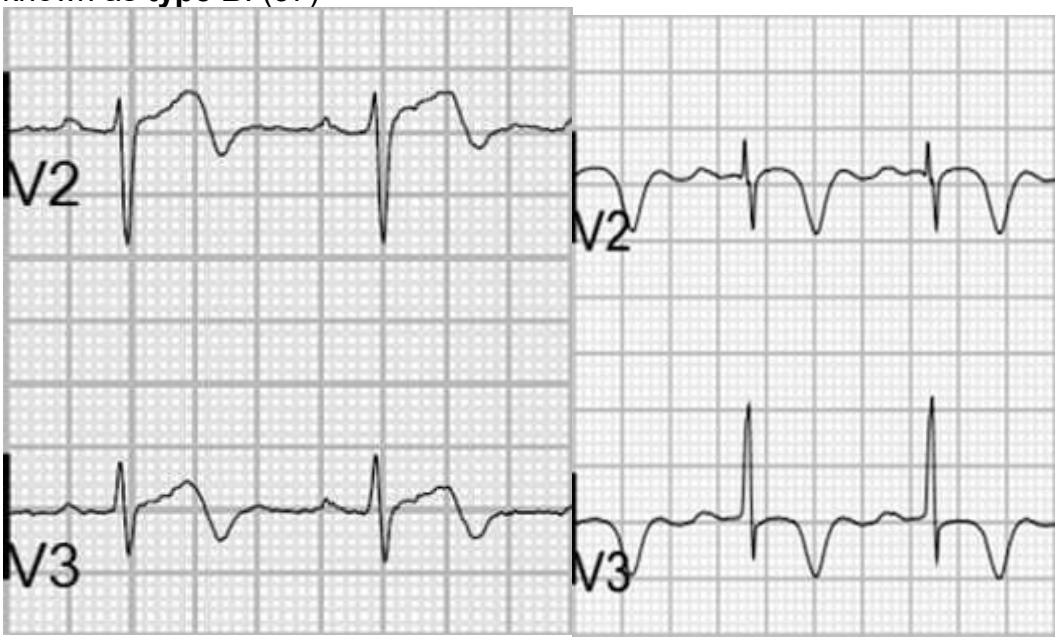


Figure 22*. Left type A Wellen's syndrome, right type B Wellen's syndrome. Image accessed from Life in the Fastlane. (37)*

These patient's may or may not have preceding chest pain. Urgent peri-operative cardiology referral is indicated.

De Winter's T waves: this recently described entity (2008) consists of upsloping ST depression in V2-6 and peaked T waves. Generally, as with other ischaemic findings, there will be reciprocal changes; this represents occlusion myocardial infarction and also requires urgent reperfusion. (37)



Figure 23. *De Winter's T waves. Image accessed from Life in the Fastlane. (37)*

Perioperative contributors to developing arrhythmias continued…

Anaesthetic factors

Medications: bradycardias may be induced by beta blockers, CCB's, digoxin and amiodarone. Acetylcholinesterase inhibitors can too, cause bradycardias if not administered with an adequate dose of an anticholinergic (atropine, glycopyrrolate). Sinus bradycardia and cardiac arrest have been documented with the administration of sugammadex. Opioids in large doses may lead to bradycardias which can be treated with anticholinergics. Vasoconstrictors may cause a temporary bradycardia by activation of the baroreceptor reflex. Effects of medication on the QT interval can be variable, those which can prolong the QT interval are depicted in figure 23 above. Local anaesthetic toxicity should remain high on the anaesthetists list of differential diagnoses and be treated appropriately with intralipid. (19)

Electrolyte disorders are a common cause of arrhythmias including deranged potassium, magnesium, and calcium. These effects are depicted below in figure 26. (19)

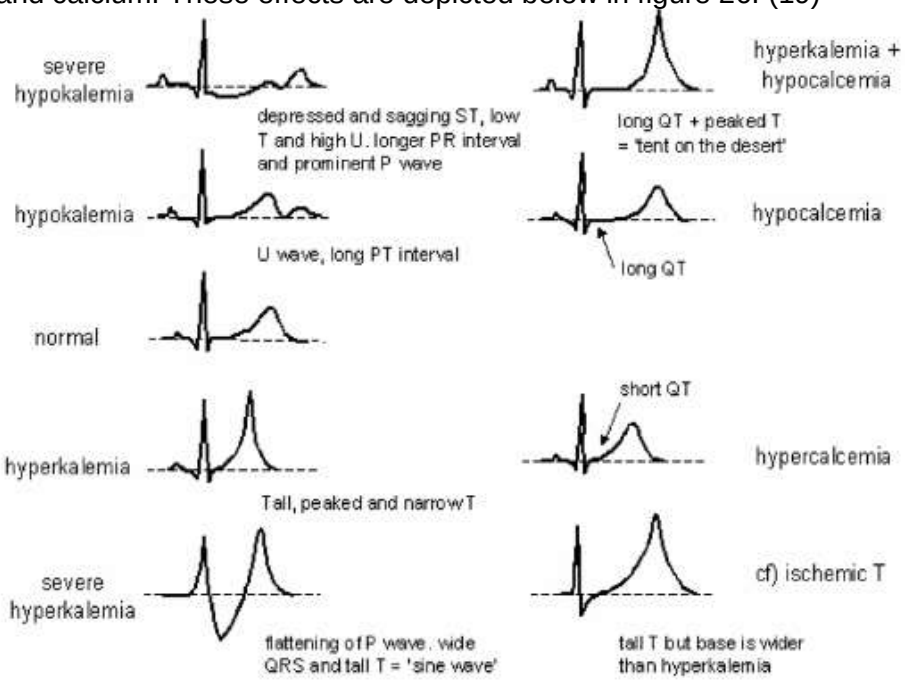


Figure 24*. ECG changes in electrolyte derangement. Image accessed from slideshare.net (38)*

Acid-base disturbances including hypoxaemia, hypocarbia and hypercarbia may cause arrhythmias. (19)

Myocardial ischaemia or heart failure predisposes the patient to conduction defects, SVT's and ectopic ventricular beats. (19)

Perioperative contributors to developing arrhythmias continued…

Surgical and interventional procedures:

Surgical procedures on the heart or in close proximity to it can cause arrhythmias. Central venous catheters can cause atrial and ventricular dysrhythmias. Patients undergoing electroconvulsive therapy may develop various arrhythmias. Peritoneal stretch, anal sphincter and uterine cervix dilatation, and extra-ocular muscle

manipulation during ophthalmological surgery can cause intra-operative bradycardias. (19)

BRADYARRHYTHMIAS
Bradycardias are classified as heart rates <60 bpm. They can be further classified by their width and regularity as follows:

- **Narrow complex**
  - **Regular:**
    - sinus bradycardia, also known as first degree heart block
    - Third degree heart block with junctional escape rhythm
    - atrial flutter with high degree block
  - **Irregular:**
    - Sinus arrhythmia
    - Mobitz 1
    - Mobitz 2
    - Atrial flutter with variable block
- **Wide complex**
  - **Regular:**
    - Third degree heart block with ventricular escape rhythm
    - Atrial flutter
  - **Irregular**:
    - Irregular bradycardia with BBB

The following figure, 27, from *uptodate* depicts intra-operative precipitants and treatment options.

Management of bradycardia in the operating room: Rapid overview*[1]

Intraoperative bradycardia with HR <60 bpm or rapidly decreasing HR with hemodynamic instability (ie, causing hypotension or poor perfusion)¶

Identify and treat potential causes of bradycardia

- Vagal reflexes
  - Initial treatment: cease surgical or interventional stimulus (eg, oculocardiac reflex during ophthalmic surgery, peritoneal stretching during laparoscopic surgery)
  - Persistent bradycardia: atropine 0.5 mg (may repeat up to a total of 3 mg) in a hemodynamically unstable patient; or glycopyrrolate 0.2 mg (may repeat up to a total of 1 mg) in a hemodynamically stable patient
- Neuraxial anesthesia with a high (ie, T1 to T4) anesthetic level
  - Initial treatment: ephedrine 5 to 10 mg or epinephrine 10 to 20 mcg
  - Persistent or severe bradycardia: larger doses of epinephrine (ie, 100 mcg) and/or continuous epinephrine infusion, as well as atropine 0.5 mg (may repeat up to a total of 3 mg)
- Medications that increase risk for sinus bradycardia (chronically or acutely administered)
  - Negative chronotropic agents (eg, beta blockers, calcium channel blockers)
    - Initial treatment: ephedrine 5 to 10 mg
    - Persistent or severe bradycardia: epinephrine 10 to 20 mg and/or continuous epinephrine infusion, as well as atropine 0.5 mg (may repeat up to a total of 3 mg)
  - Anticholinesterase agents, opioids, vasoconstrictors
    - Initial treatment: glycopyrrolate 0.2 mg (may repeat up to a total of 1 mg)
    - Persistent or severe bradycardia: atropine 0.5 mg (may repeat up to a total of 3 mg)
- Auto-PEEP or high peak inspiratory pressures
  - Initial treatment: hand ventilation with attention to airway pressures
  - Persistent or severe bradycardia: atropine 0.5 mg (may repeat up to a total of 3 mg)
- Less common causes of bradycardia: identify and treat as needed:
  - "H's": hydrogen ion (ie, acidemia), hypoglycemia, hypokalemia, hyperkalemia, hypermagnesemia, hypothermia, hyperthermia (ie, malignant hyperthermia)
  - "T's": tamponade, tension pneumothorax
  - Myocardial ischemia
  - Local anesthetic systemic toxicity (LAST)
  - Exacerbation of pulmonary hypertension

Persistent bradycardia that does not respond to pharmacologic treatment

- Temporary pacing
  - Initial treatment: transcutaneous pacing
  - Persistent bradycardia: prepare for transvenous pacing (ie, insertion of a pacing lead or pacing PA catheter) via central venous access
- Expert cardiology consultation
- Transfer to intensive care unit for postoperative management

Severe or persistent bradycardia progressing to asystole, pulseless electrical activity (PEA)

- Begin CPR and ACLS and call for help (in addition to treatments listed above)Δ
- Consider other therapies◊

HR: heart rate; bpm: beats per minute; PEEP: positive end-expiratory pressure; PA: pulmonary artery; CPR: cardiopulmonary resuscitation; ACLS: advanced cardiac life support.

* Refer to UpToDate content discussing arrhythmias during anesthesia and perioperative ACLS.

¶ A HR of 40 to 60 bpm without symptoms is common during general anesthesia, and does not require treatment.

Δ Refer to UpToDate content discussing basic life support and ACLS.

◊ Refer to UpToDate content discussing therapies of uncertain benefit in basic and advanced cardiac life support.

Reference:
1. Mishra VK, Ehsan S, Thau K-C, et al. Cardiac arrest in the operating room: Resuscitation and management for the anesthesiologist. Anesth Analg 2018; 126:876.

UpToDate

***Figure 25**. Management of bradycardia in the operating room. Image accessed from uptodate. (18)*

The American Heart Association 2020 guidelines are useful for unstable, bradycardic patients.

See Figure 28 below.

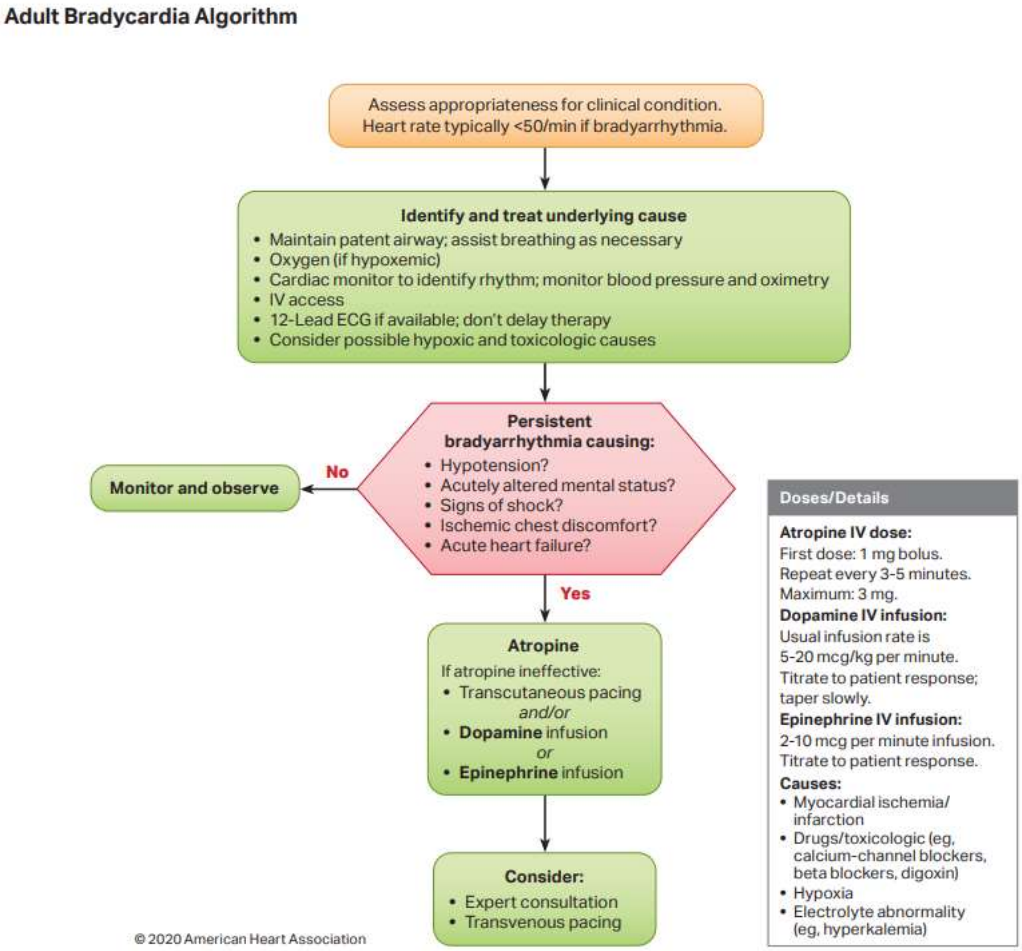


Figure 26. *AHA 2020 bradycardia management algorithm. (39)*

Symptomatic bradycardic patients will require electrical or chemical pacing. Atropine is the first line of treatment. The dose has been adjusted from 0.5mg to 1mg in adults. If repeat doses of atropine are required a dopamine or adrenaline infusion or transcutaneous pacing can be considered.

Anaesthetic related causes of bradycardia are related to vagal stimulation which may occur with IV cannulation, epidural anaesthesia and suctioning. Neurocardiac syncope may occur 30 minutes after administration of a spinal anaesthetic precipitating unexpected bradycardic arrest. Common causes of bradycardias including electrolyte abnormalities should be sought. A unique category of bradycardia may develop in the transplanted heart. These hearts are devoid of innervation and thus will not respond to anticholinergic medication; in this scenario chronotropic agents such as isoproterenol or adrenaline can be used. (18)

Surgical causes of bradycardias are mostly vagal stimulation by peritoneal stretch and anal or cervical dilatation may precipitate bradycardias. Other notable anaesthetic related causes of bradycardias include the oculocardiac reflex during ophthamological surgery and carotid plaque excision during endarterectomy.

Stable anaesthetic or surgical induced bradycardias can be treated with 0.2mg of glycopyrrolate to moderately increase the HR but avoid a severe tachycardia. The anaesthetist should familiarise themselves with transcutaneous pacing, whereby pacing pads are placed across the chest and an electrical stimulus is applied to cause ventricular myocyte depolarisation and contraction. Successful pacing is demonstrated by electrical (ECG QRS), mechanical (palpable pulse) and physiological capture (improvement in chest pain or level of consciousness). Cardiology consultation may be required if transvenous pacing is required, such as in situations where access to transthoracic pacing is not possible due to surgical access. (18)

SPECIFIC BRADYARRHYTHMIAS
(see figure 28)

Second degree heart (AV) block
This bradycardia is characterised by a prolonged PR interval > 200msec with intermittent non-conducted p waves or dropped beats, these can be subclassified as follows:

- Mobitz type 1 AV block: constant PR prolongation and dropped beat. These patients can be monitored and generally do not require treatment unless symptomatic.
- Mobitz type 2 AV block: progressive lengthening of the PR interval with dropped beat. This arrhythmia is prone to progressing to a complete heart block and thus peri-operative referral for permanent pacing is prudent. (18)

Third degree heart (AV) block
Atrial cardiac conduction ceases to be conducted through the AV node. There is complete discordance between p waves and QRS complexes. The intrinsic heart rate in complete heart block is generally 30-40 bpm. The QRS can either be narrow, as seen with junctional depolarisation or wide, as seen with ventricular depolarisation, depending upon the origin of the spontaneously depolarising tissue. (18) Acute perioperative identification requires pacing of some sort – the urgency of which, determined by acuity of surgery and haemodynamic status of the patient. This will likely determine whether the pacing is transthoracic, chemical, or temporary venous pacing; Permanent pacing may also be inserted.

Sinus node dysfunction
A diseased SA node may produce irregular conduction of cardiac impulses. The patient may present with long pauses between QRS complexes, junctional escape rhythms or sinus arrest. Searching for the underlying cause and treatment according to the AHA recommendations is recommended. (18)

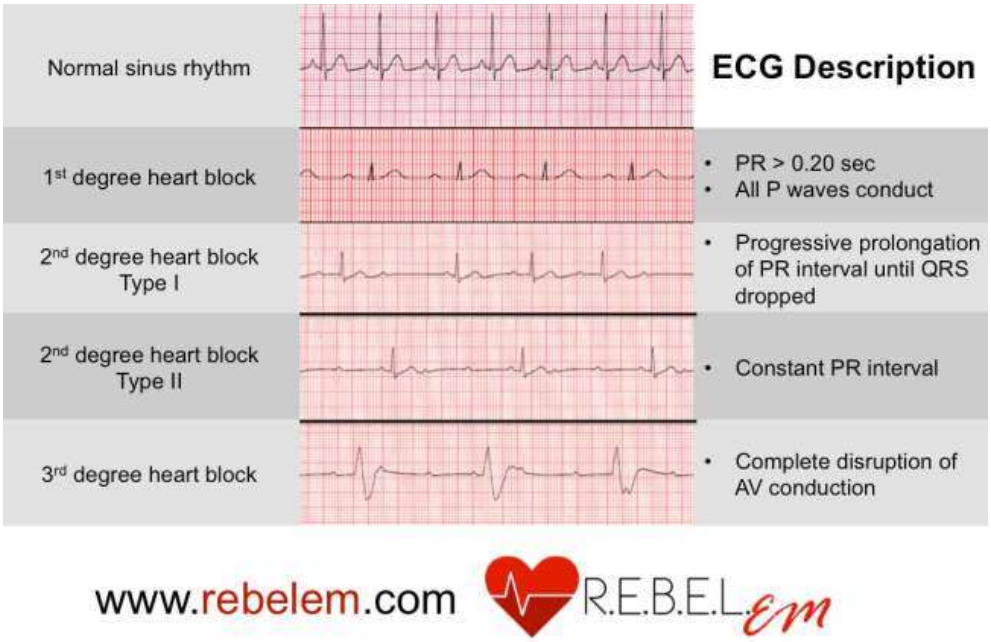


Figure 27: *Bradycardia ECG rhythm strips, 3rd degree AV block with wide complex QRS denoting ventricular escape rhythm accessed from rebelem.com. (40)*

TACHYARRHYTHMIAS

Tachyarrhythmias are defined as heart rates > 100 bpm. Like bradyarrhythmias, tachyarrhythmias can be classified according to width and regularity:

- **Narrow complex tachycardia**
  - **Regular**
    - Sinus tachycardia
    - Atrial flutter
    - AVRT
    - AVNRT
  - **Irregular**
    - Atrial fibrillation
    - Atrial flutter with variable block
    - Multifocal atrial tachycardia
- **Broad complex tachycardia**
  - **Regular**
    - VT
    - SVT + BBB
    - SVT with aberrant conduction
    - SVT with eccentric pacing
  - **Irregular**
    - VF
    - AF + BBB
    - AF + aberrant conduction
    - Torsades Du Pont

Basic management principles of management of tachycardias

Management principles as suggested by the American Heart Association guidelines from 2020.

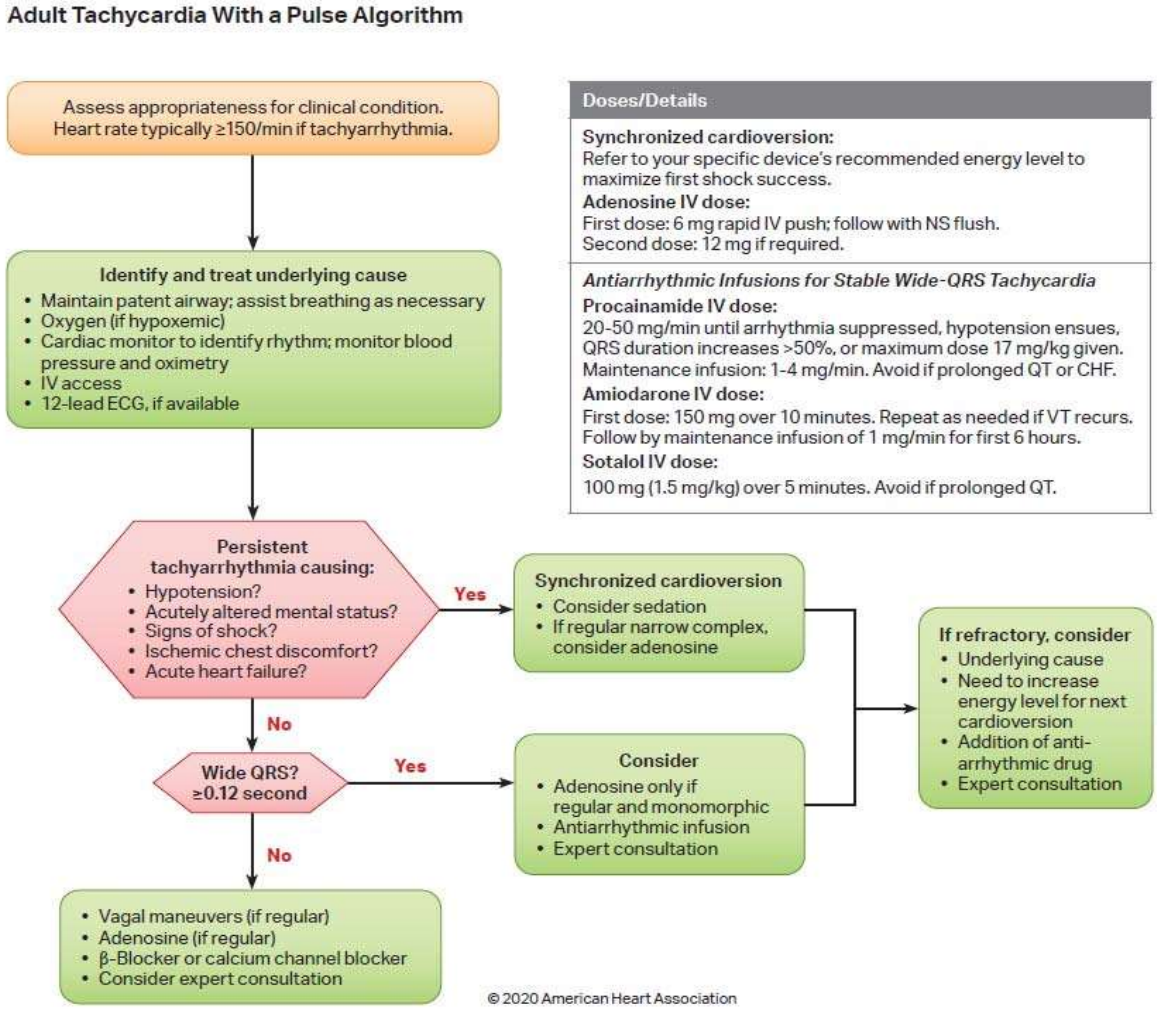


Figure 28. *AHA tachycardia with a pulse algorithm. (4)*

Specific tachyarrhythmias

Narrow and regular: Sinus tachycardia

Although relatively benign this dysrhythmia can be associated with ST depression indicative of ischaemia. The most common intraoperative causes are pain, sepsis, fever and hypovolaemia. Clinicians should keep thyrotoxicosis and pulmonary embolism in mind. ECG findings are a HR > 100 bpm with a p wave visible in front of every QRS. (3) The PR interval is usually < 200msec, if > 200msec suspect atrial tachycardia. (21). Treatment aims at correcting the underlying cause. Other treatment considerations include short acting IV Beta Blockers (Esmolol 20 – 50mg, or 0.5mg/kg, boluses every 2 to 3 minutes) in patients at risk of ischaemic heart disease or patients with fixed cardiac output valvular lesions e.g., Mitral/Aortic stenosis. Esmolol infusions can be considered if bolus doses are effective. If >200mg of Esmolol are required without cessation of the tachycardia an alternative diagnosis should be sought. Exercise caution with Beta blockers in hypovolaemic patients, patients with heart

blocks and asthmatics. Labetalol (5-10mg) is a reasonable alternative if Esmolol is not available however, it is longer acting and non-selective. (18)



Figure 29. *Sinus tachycardia. Image accessed from uptodate.com. (18)*

The image below shows another useful schematic representation of the classifications of narrow complex supraventricular tachycardias:

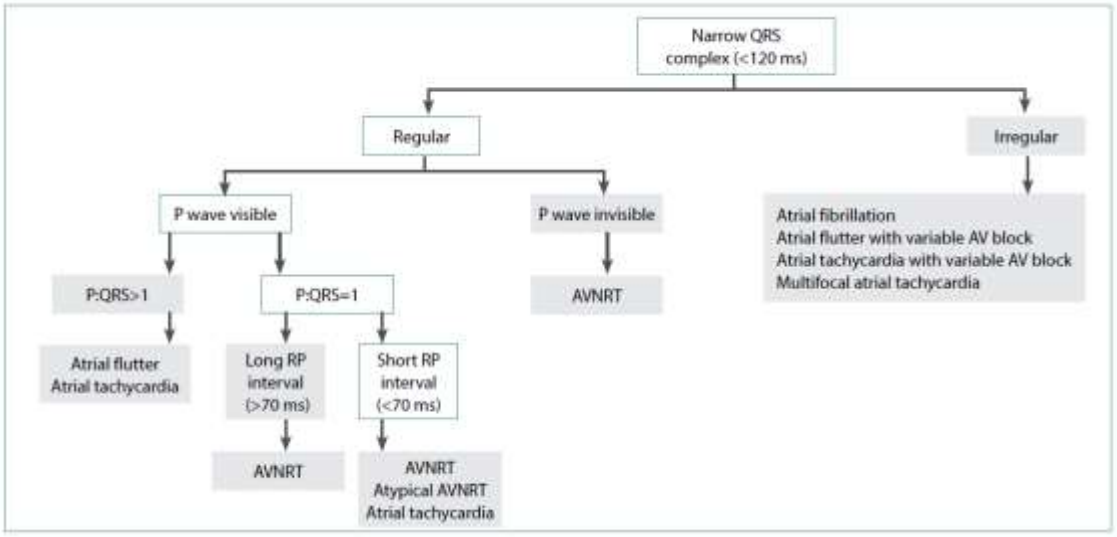


Figure 30. *Classification of SVT's. Image accessed from the South African Journal of Anaesthesia. (41)*

In haemodynamically stable patients the goals are firstly to attempt to convert supraventricular arrhythmias. Simple manoeuvres are employed which include vagal stimulation (Valsalva and carotid massage). Adenosine can be considered starting at 6mg followed by a 12mg saline flush. The 12mg dose can be repeated. **If the SVT is dependent on the AV node the arrhythmia will cease after adenosine; if the SVT is independent of the AV node it will continue. (5)** The second aim is to control the ventricular rate. The aim of treatment is to target the AV node to prevent a fast ventricular rate. Agents with class II (Beta blockers) or IV (calcium channel blockers) activity are used. Esmolol is a suggested Beta Blocker and Verapamil and diltiazem are suggested calcium channel blockers. (refer to doses under Vaughn Williams classification above). Cardioversion can be considered in patients who don't respond to IV medication. (41)

Narrow and regular tachyarrhythmia: Atrial flutter

The origin of atrial flutter is usually a re-entry circuit in the right atrium. The average atrial rate is 300bpm. The ventricular rate depends on how many atrial beats are conducted through the AV node. In a 2:1 block with the atrial rate at 300bpm the ventricular rate will be 150bpm. In a 3:1 block the ventricular rate will be 100bpm, in a 4:1 block the ventricular rate will be 75bpm. (21) If the patient is unstable the ALCS algorithm should be followed; in stable patients the management is similar to atrial

fibrillation (discussed below). (18) ECG findings include a narrow complex, regular tachycardia with no discernible p waves. A typical “saw tooth” pattern is visible. (see figure 32)

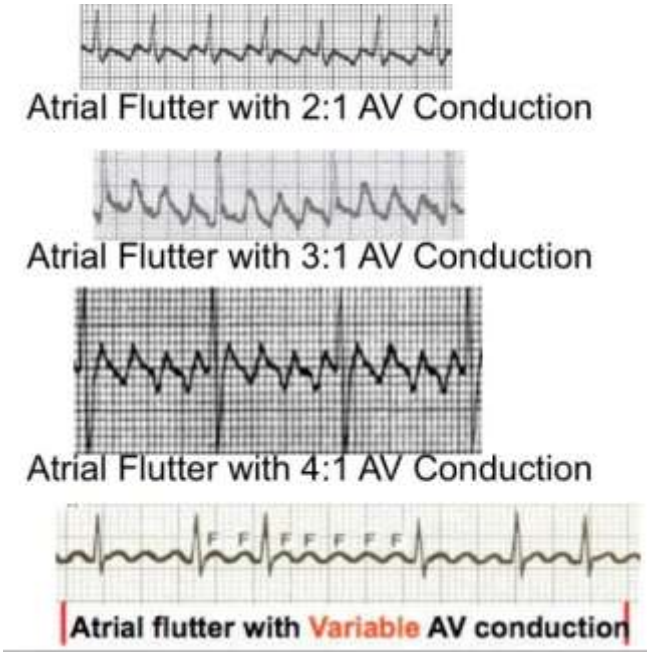


Figure 31 *(right). Atrial flutter. Image accessed from www.unm.edu (42)*

Narrow and regular tachyarrhythmia:
Atrioventricular re-entry tachycardia (AVRT)
These patients have an accessory pathway; the re-entry circuit usually occurs between the AV node and accessory pathway. The circuit may be directed in an *orthodromic* or *antidromic* direction. (1) Accessory pathways may present as a shortened PR interval on a pre-operative ECG as seen in Wolf-Parkinson White syndrome and Lown-Ganong-Levine (LGL) Syndrome.
In orthodromic conduction the circuit travels in the usual fashion, anterograde, via the AV node thus producing a narrow complex tachycardia. The exception is if there is a pre-existing BBB which would produce a wide complex rhythm. ECG features: rate 200-300bpm, retrograde p waves.

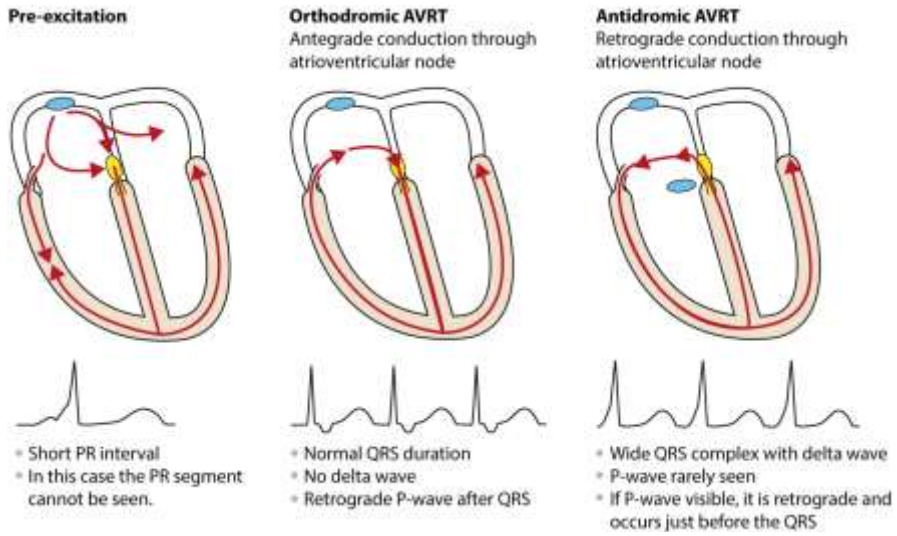


***Figure 33**: orthodromic and antidromic conduction pathways in AVRT. Image accessed from www.ecgwaves.com (2)*



***Figure 32:** Orthodromic AVRT will produce a narrow complex SVT (top picture). Antidromic AVRT will produce an irregular wide complex SVT, thus, AHA recommends adenosine only of regular (i.e., **not** antidromic AVRT), starting **certain** antiarrhythmics and expert consultation. (Bottom picture). Unstable patients should receive synchronised cardioversion. (4) Image accessed from Life in the Fastlane. (1)*

In antidromic conduction (remember *anti*-clockwise) the circuit travels in a retrograde direction, first activating the ventricular tissue and then the atrial tissue, thus producing a wide complex tachycardia, usually at a rate of 200-300bpm.

Important: if AV nodal blocking agents (Adenosine, Digoxin, Beta blockers or CCB's) are administered in these patients, especially in antidromic AVRT there is a small chance of promoting the circuit down the accessory pathway and thus precipitating cardiac arrest or an irregular broad complex tachycardia as the ventricle will contract at 200-300bpm. Procainamide, a VW class 1a antiarrhythmic is an attractive treatment option as it decreases the excitability of non -nodal tissues and thus slows conduction over the accessory bundle. Flecainide (class 1C) and amiodarone (class III) may also be considered. (43)

Definitive treatment involves ablating the accessory pathway. (44)

Narrow and regular tachyarrhythmia: Atrioventricular nodal re-entry tachycardia (AVNRT)

This SVT is the most common cause of paroxysmal SVT's. Patients have a re-entry circuit in their AV node and generally have structurally normal hearts. (44)
Patients may present intra-operatively as stable or unstable. Haemodynamic instability will dictate the need for immediate cardioversion or not.

ECG findings: narrow complex, regular tachycardia with p waves hidden in QRS (usually not visible). Rate typically 150-250bpm. (44)

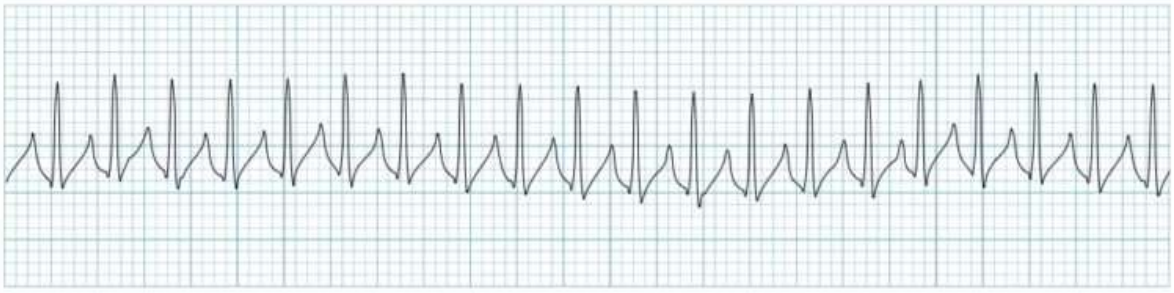


Figure 33: *An SVT likely caused by AVNRT or AVRT. Image accessed from Life In The Fastlane. (1)(5)*

Irregular narrow complex tachycardias

Atrial fibrillation (AF): This is a very common and important arrhythmia for the anaesthetist to know about. It occurs in 5-10% of patients over the age of 65 years. The arrhythmia develops as there are multiple areas of atrial myocardium in different phases of depolarisation and repolarisation. The fibrillating atria can depolarise at 300 – 500 times a second. A functioning AV node will effectively filter the impulses to the ventricles at a rate less than 200bpm. A blocked AV node, or an AV node with an accessory pathway as seen in AVRT can precipitate very fast, irregular tachycardias. (44) AF may occur pre-operatively, abruptly intra-operatively or post-operatively. (18). There are multiple precipitants which ought to be sought, including thyrotoxicosis, electrolyte disturbances, sepsis, valvular heart disease, hypertensive heart disease ischaemic heart disease, chronic lung disease and cardiomyopathies, amongst others.

(44) Pre-operatively, patients with AF should be have electrolytes checked. Potassium serum levels > 4mmol/l and magnesium serum levels >1mmol/l should be aimed for.

Treatment goals in AF are to prevent thromboembolism with anticoagulation, this may prove tricky in the peri-operative setting, rate control, identify the underlying cause and to regain sinus rhythm if possible. (44) Haemodynamically unstable patients with fast AF require immediate synchronised cardioversion. (18) Haemodynamically stable patients with fast AF are at risk of venous thromboembolic disease if the AF is known to be chronic; cardioversion may precipitate embolism and pharmacotherapy is preferred here. Beta blockers and calcium channel blockers can be used here to slow the ventricular rate. (18) AF with HR <120 bpm is generally haemodynamically stable; patients in this group, who need rate control due to underlying ischaemic heart or stenotic can be started on a beta blocker or CCB, provided no contraindications such as left ventricular failure, asthma or AVRT co-exist. Amiodarone can be used to slow the ventricular rate and for maintenance therapy. (18) Alternative therapies include Digoxin and Flecainide. (45)

In elective patients with suspected chronic AF who remain haemodynamically stable should be referred appropriately, anticoagulated and then rate controlled. (45)

The AFFIRM trial demonstrated that rate control is superior to rhythm control. (46)

The cessation of anticoagulation preoperatively should be based on the patient's underlying risk of thromboembolisation. High risk patients with CHA2DS2-VASc score greater or equal to 7 will likely benefit from anticoagulation bridging. In patients with a moderate risk of VTE (CHA2DS2-VASc 5-6) bleeding risks and VTE risks need to be balanced. Lower risk VTE patients (CHA2DS2-VASc less than or equal to 4) may safely interrupt anticoagulation for longer periods. (47)



***Figure 34**: Atrial fibrillation, irregular narrow complex tachycardia with no discernible p waves visible. Image accessed from Life in the Fastlane. (5)*

Multifocal atrial tachycardia (MAT)

This narrow complex irregular tachycardia originates from multiple foci in the atrium. Characteristic ECG findings are differing p wave morphologies. Patients generally have abnormal underlying pulmonary disease. If stable perioperative rate control may be required. Unstable patients may require cardioversion. If identified electively surgery should be postponed until after cardiology consultation. (19)

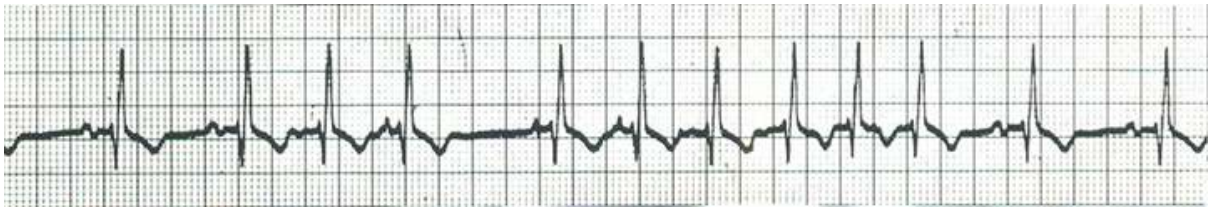


Figure 35. *Multifocal atrial tachycardia. Image accessed on slideshare.net. (19)*

WIDE COMPLEX TACHYCARDIAS

Wide complex tachycardias imply that the origin of depolarisation is either ventricular in nature or is transmitted via an aberrant pathway.

Unstable patients should receive synchronised cardioversion, or, if pulseless: CPR and defibrillation. Synchronised cardioversion is performed if the T wave and QRS complex can be seen; if they are not visible defibrillation should be attempted (biphasic 120-200J; monophasic 360J).(48) The South African Resuscitation council recommends starting at 100J synchronised cardioversion whether using a biphasic or monophasic defibrillator. (48, 49)

In patients who have stable, broad tachycardias a 12 lead ECG can be attained to try and determine the underlying aetiology.

In haemodynamically stable patients with a wide, monomorphic tachycardia ("monomorphic regular broad tachycardia" refers to VT; the QRS complexes throughout the ECG look similar, whereas "polymorphic" refers to beat to beat variation in QRS morphology)(50). **The treatment approach for stable VT, also known as monomorphic wide tachycardia, is as follows:**

- Attempt vagal manoeuvres and give adenosine: this will abort any nodal causes of the tachycardia (AVRT, AVNRT). AF and atrial flutter may be unmasked by adenosine. Note that resuscitation equipment should be available as adenosine may cause cardiovascular collapse.

- Using the response to vagal manoeuvres as a diagnostic aid: sinus tachycardias will temporarily slow, as will AF or atrial flutter; the underlying atrial depolarisation will remain the same, however. Paroxysmal SVT's will terminate (AVRT/AVNRT) due to dependence on the AV node. **VT will not respond to adenosine or vagal manoeuvres** thus these manoeuvres assist with the diagnosis. CCB's, beta-blockers and digoxin are not recommended at this stage as they may cause hemodynamic deterioration. (4, 48).

- Adenosine may convert SVTs to AF.

- A cautionary note, as previously mentioned: The wide complex tachycardia may be due to underlying pre-excitation syndromes such as WPW. Adenosine will cause transient AV node blockade with propagation down the accessory pathway; if the accessory pathway is antidromic, this can cause ventricular depolarisation at 200-300bpm leading to cardiac arrest. A defibrillator should thus be available. (48)

Keen readers are encouraged to visit *VT versus SVT: It's as easy as ABCDE* • LITFL to help distinguish between SVT's and VT. Notably, the most accurate sign found on ECG to distinguish the two is concordance (i.e., if concordant, the complexes are either predominantly positive or negative: lack of RS complexes in the precordial leads (V1-6) is 97% specific for VT. Thus, if the complexes are **all** positive or **all** negative, this suggests VT. Other specific findings are that VT is the most common diagnosis of wide complex tachycardias (80%) and that a broad complex >200ms favours VT. (51) See figures 40 and 41 below

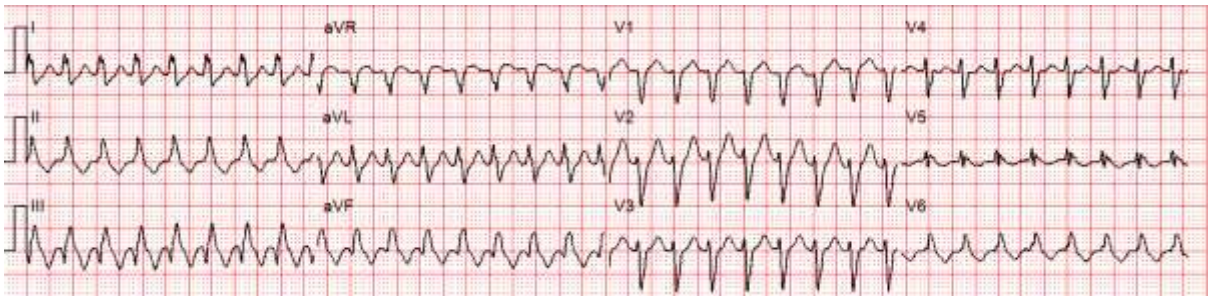


Figure 36. *Wide complex tachycardia with discordance suggesting SVT with aberrancy. Image accessed from aclsmedicaltraining.com (52)*

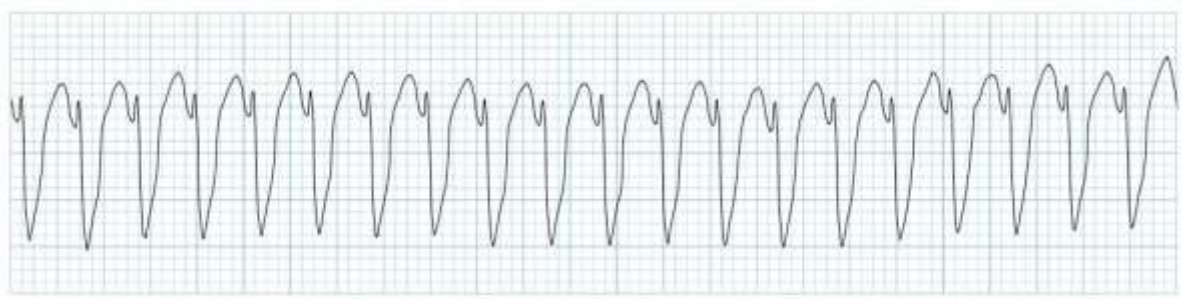


Figure 37. *Monomorphic VT. Image accessed from Life in the Fastlane. (51)*

Once the diagnosis of VT has been made the patient can either be cardioverted, defibrillated or started on amiodarone, depending on the clinical situation. Amiodarone is administered as 150 mg IV in 5% dextrose over 10 minutes and then 360mg over 6 hours and 540mg over 18 hours. Procainamide is advocated in international literature and by the AHA however, it is not available in the KZN public sector. (4, 19). It must be remembered that electrolytes and underlying conditions e.g., ischaemia must be corrected.

AHA tachycardia algorithm versus SA resus council algorithm:
The South African Resuscitation Council algorithm is similar to the AHA algorithm. Differences arise in the wide complex QRS tachycardia; the AHA advocates the use of procainamide and sotalol, whereas the South African Resuscitation Council suggest only amiodarone. (49)

PERIOPERATIVE CARE AND INTERVENTIONAL PROCEDURES
Monitoring: the type of anticipated arrhythmia and haemodynamic sequelae thereof should be identified pre-operatively; invasive arterial blood pressure monitoring and a 5 lead ECG may be prudent to use in cases where haemodynamic collapse may occur.

Postoperative patients should be managed according to the nature of their arrhythmia and may require ICU or high care. Ongoing, underlying arrhythmias should be referred to cardiology and patients with benign, uncertain arrhythmias should be seen by a physician before discharge.

Electrophysiological ablation: this is recommended for certain arrhythmias.

This booklet does not cover this, but knowledge thereof is recommended.

CONCLUSION

Peri-operative arrhythmias are a common entity varying from benign to life threatening. The subtleties and intricacies around diagnosis and treatment of them is not often portrayed. Although, not completely inclusive of all arrhythmias this booklet aims at helping the anaesthetist in South Africa diagnose and treat some of the peri-operative arrhythmias they may encounter with medication available to them.
Having a basic approach to the arrhythmias is key to treating them; excluding common causes and then troubleshooting will enable the anaesthetist to achieve this.

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